Kakebe

PROPOSAL

Digital Platform for the Lango Cultural Foundation

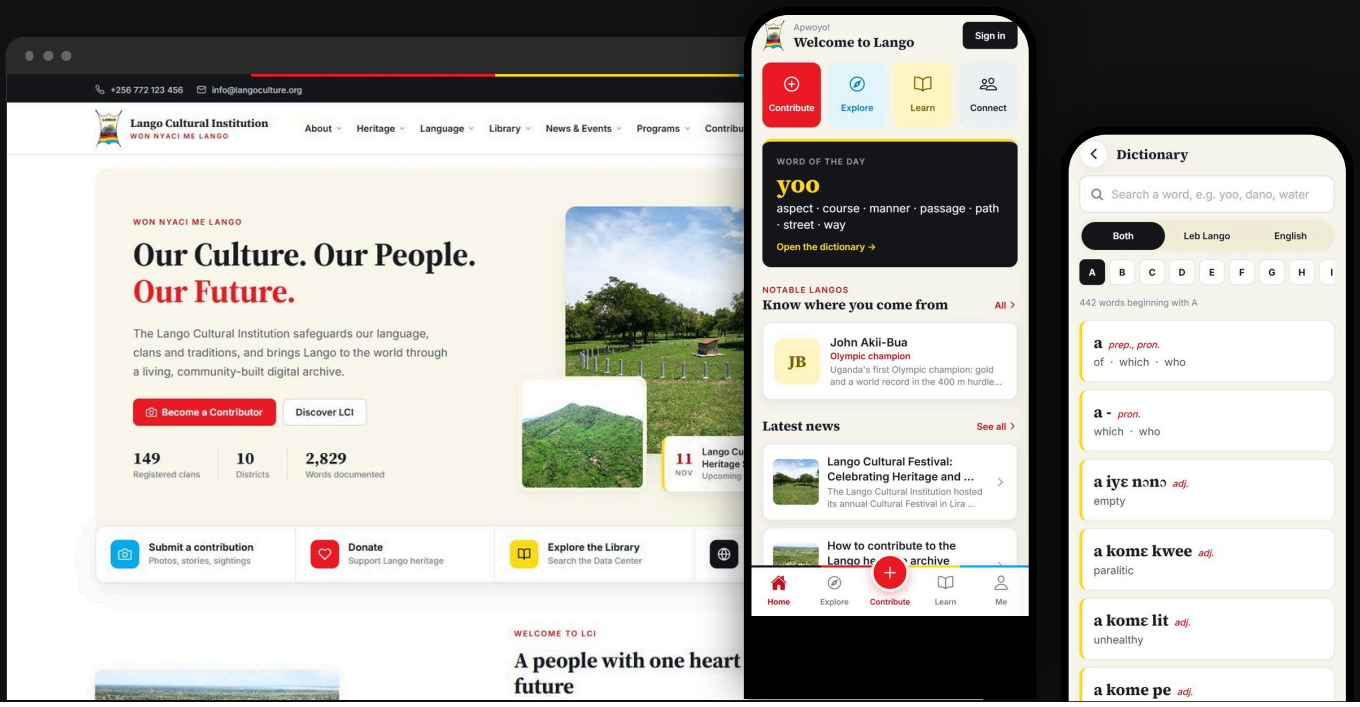
Website · Admin Control Panel · Lango Mobile App

PREPARED FOR
Lango Cultural Foundation

PREPARED BY
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Lira City, Uganda

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About this document

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How to read this proposal

- **Decision makers** can read Chapter 1 (Executive Summary), Chapter 3 (Our Recommendation) and Chapter 13 (Investment).
- **Programme and content teams** will find the detail in Chapters 4 to 7 and the full feature catalogue in Appendix A.
- **Technical and legal reviewers** will find architecture, security and data protection in Chapter 8, and assumptions and terms in Chapter 17.
- **Evidence of delivery** is in Chapters 9 and 10, and the working demonstrator is shown in Appendix B.

Conventions

- Text like [TBC: item] marks something Kakebe or the Foundation must confirm or complete before the proposal is issued as final.
- In Appendix A, ● means already working in the demonstrator, ◐ means built in this engagement, and ○ means optional or roadmap.
- Amounts are in Ugandan shillings (UGX) and exclude taxes unless stated. Timelines are indicative until confirmed at the end of Phase 0.

Document history

Version	Date	Author	Change
1.0	26 October 2026	Kakebe Technologies	First draft for discussion

Confidentiality

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CHAPTER 1

Executive Summary

A living, searchable, offline-ready home for Lango language, clans, history and heritage: one platform, delivered by a team that already built a working version.

Lango Cultural Foundation is the keeper of a heritage that is older than any archive: the Leb Lango language, a clan system that runs to well over a hundred clans, the oral history of nine districts and Lira City, and the ceremonies, songs, crafts and proverbs that tie them together. Much of that knowledge still lives in the memory of elders, in family albums, in scattered documents and in WhatsApp groups. Every year some of it is lost. At the same time, a young, phone-first and diaspora-connected generation expects to find its culture where it already spends its time: on a screen in its pocket.

Kakebe Technologies Limited proposes to close that gap with **one digital platform made of three connected parts**:

Part	What it is	Who it serves
The public website	A fast, bilingual, brand-led home for history, districts, clans, notable Langos, heritage, the Language Hub (dictionary, translator, proverbs, lessons), news, events, programmes and giving	Everyone: citizens, students, researchers, diaspora, partners, media
The admin control panel	A role-based workspace where the Foundation's own staff publish, review, translate and report, with no developer needed for day-to-day work	Editors, moderators, language experts, clan leaders, administrators
The Lango Mobile App	Android, iOS and installable web app with the offline dictionary, guided contribution forms (GPS, camera, voice), notifications and member recognition	Youth, clan members, contributors in the field, the diaspora

All three share **one backend, one database and one review queue**. A word suggested on a phone in Aduku appears in the same moderation screen as a word suggested on a laptop in London; once an expert approves it, it is live in the dictionary and the translator on the website and in the app at the same moment.

1.1 Our recommendation

> **We recommend building the website and the mobile app together, on a single shared platform (Option B in Chapter 13).**

The reasons are practical, not fashionable:

1. **Reach.** The website is how the world finds the Foundation through search engines, shared links and press. The app is how the community uses it every day: offline, with the camera and microphone, with a push notification when a contribution is approved.
2. **Cost.** The app and the website use the same API, accounts, content and moderation. Building them together costs far less than building them one after the other, and it avoids two sources of truth.
3. **Capture.** The most valuable thing the Foundation can collect is knowledge that is not yet written down. That happens in villages, at ceremonies and in elders' homes, often without a data connection. Only an offline-first app can capture it there.

4. **Continuity.** Content, accounts and contributors are built once and reused for every future channel, whether that is SMS and USSD, a community forum or a radio programme.

1.2 This is not a concept: a working version already exists

Kakebe has already built and tested a complete working demonstrator of this platform for the Lango community (built under the working title *Lango Cultural Institution*). It is not a slide deck. You can log in to it today, and the scope in this proposal is measured against what it already does:

What exists today	Measure
Public website	More than 30 page types: history, districts, clans, people, heritage, language, library, news, events, programmes, giving and more
Admin control panel	30 managed content types plus bespoke screens for moderation, translations, notifications, users, audit log and backups
Mobile app	Around 50 screens on one codebase for Android, iOS and the installable web app; offline dictionary; offline contribution queue
Back end	A documented JSON API, a database of 32 tables, token-based sign-in, rate limiting and duplicate-safe uploads
Content rules	Every fact is sourced; nothing cultural is invented; the interface is built to receive the Foundation's own experts' words

Starting from a working system changes the risk profile of the whole project. **Our first weeks are spent adapting, extending and filling the platform with the Foundation's authority, not discovering what a cultural platform is.** Appendix B shows the demonstrator screen by screen.

1.3 What the Foundation gains

- **A trusted public record** of the language, clans, districts, history and heritage, with sources shown and reviewers named.
- **A Language Hub** that teaches Leb Lango and makes it searchable online and offline, growing through community contributions that experts approve.
- **A community contribution engine** with tracking codes, GPS-tagged observations, photographs, audio and documents, reviewed in one queue.
- **A mobile presence** that works in low-connectivity areas and keeps the diaspora close.
- **Transparent giving and programmes:** pledges, programme progress and transparency figures that build donor confidence.
- **Independence:** an admin team trained to run the platform, full documentation, no lock-in, and ownership of the content and data.
- **A youth skills pipeline,** because the platform is built, maintained and populated in Lira City, with trainees from Kakebe's own programmes working under supervision.

1.4 Why Kakebe Technologies

Kakebe is a youth-led technology company based in **Lira City**, in the heart of Lango. Our portfolio spans non-profit foundations, a city government website, a ministry scheduling platform, a national boda-boda rider registry, church and community apps with mobile money, crowdfunding, event ticketing, point of sale, e-learning and sign-language education (Chapter 10). We already operate the exact technologies this project

needs: content-managed websites, multi-role admin systems, Expo/React Native mobile apps backed by a shared API, offline-first design, and mobile-money payment integration. We have trained hundreds of young people (257 graduated from Tech Camp 2025 alone), and we intend this project to leave Lango with more skills as well as a platform.

1.5 The proposal at a glance

Item	Detail
Recommended scope	Public website + admin control panel + Lango Mobile App (Android, iOS, web app)
Indicative delivery	About 20 weeks from signed agreement to public launch, followed by 90 days of included hypercare (Chapter 11)
Investment	[TBC: total one-time investment in UGX for Option B], with a transparent breakdown and milestone payments (Chapter 13)
Hosting and running costs	Itemised annually; Kakebe can manage hosting or hand it to the Foundation (Chapters 8 and 13)
Languages	English and Leb Lango throughout; interface wording entered by the Foundation's language experts, never machine-translated
Ownership	Content, data and accounts belong to the Foundation; documentation and handover included
Support	Three support tiers with defined response times (Chapter 14)

1.6 The next step

We would like to hold a **discovery workshop of about 90 minutes** with the Foundation's leadership and a small working group of language experts, historians and communications staff. At the end of it we will confirm scope, finalise the quotation and agree a start date. Chapter 18 sets out the steps, and the contact details for the Kakebe team are on the last page.

CHAPTER 2

Understanding the Foundation's Needs

What the platform must achieve, the challenges it must overcome, and the principles that guide every decision in this proposal.

2.1 Context

The Lango people live in the Lango sub-region of northern Uganda, an area of nine districts (Lira, Apac, Oyam, Kole, Dokolo, Amolatar, Alebtong, Otuke and Kwania) and Lira City, home to more than two million people according to the 2024 census. Lira City is the commercial and educational centre of the region. The Lango language, Leb Lango, is spoken in homes, markets and on the radio; English and Kiswahili are used in offices and business.

Lango Cultural Foundation is the cultural institution that carries the identity of that community: its traditional leadership under the Won Nyaci, its clans, its ceremonies, its language and its history. A digital platform for such an institution is therefore not a marketing website. It is a **custodial** system, one that holds knowledge in trust, shows where each fact came from, and keeps authority over what is published firmly in the hands of the people the community recognises.

> **How we have written this proposal.** We describe the Foundation's needs as we understand them from public information and from the work we have already done on the demonstrator. Statements marked **[TBC]** are points to confirm with the Foundation at the discovery workshop. We would rather ask than assume.

2.2 Objectives we believe the platform should serve

#	Objective	What success looks like
1	**Preserve**	Language, clans, history and heritage are recorded in a durable, backed-up, searchable archive with sources
2	**Teach**	A young Lango, or a child abroad, can learn words, proverbs and greetings and hear how they are said
3	**Connect**	Clans, districts and the diaspora can find each other, take part and hear from the Foundation
4	**Represent**	The Foundation has an authoritative public voice and a trusted reference that media, schools and researchers can cite
5	**Sustain**	Programmes, events and giving are visible and accountable, supporting the Foundation's work
6	**Include**	The platform works for elders and youth, in town and village, online and offline
7	**Govern**	Staff can run it themselves, with clear roles, review and an audit trail

These objectives are the yardstick we use throughout. Each feature in Chapters 4 to 6 exists to serve at least one of them, and Chapter 16 turns them into measurable indicators.

2.3 The challenges the platform must solve

Challenge	Consequence if unaddressed	How the platform responds
Knowledge is held in people, not on paper.	When an elder passes away, a clan's oral history, songs and proverbs can go with them.	Guided contribution forms for oral history, proverbs, words, heritage sites and documents, with audio recording, photos and GPS, reviewed and archived.
Language shift among youth.	Fewer young people speak or write Leb Lango confidently.	Dictionary, English to Leb Lango translator, proverbs, greetings, phrasebook and lessons, usable offline on a phone.
A dispersed community.	Clans, nine districts and a large diaspora are hard to reach through one channel.	Website for reach and search, app for daily use, a diaspora corner, clan pages and a clan picker on every registration.
Patchy and expensive data.	A web-only platform excludes the people who most need it.	Offline-first app, versioned offline dictionary, stale-while-revalidate caching, queued uploads, lightweight pages.
Concern about authenticity.	Wrong or invented content would damage trust in the whole institution.	A strict content integrity policy: sources displayed, status labels, expert review, and nothing invented (Chapter 7 and Appendix C).
Paper-based or ad hoc processes.	Registrations, applications and event tickets are slow to process and hard to report on.	Online forms with tracking codes, a unified applications inbox, event registration with printable tickets.
Limited IT capacity.	A platform that needs a developer for every change will stall.	A no-code admin, role-based access, training, documentation and a support plan.
Sensitive public matters.	Leadership and other sensitive topics can be contested and must be handled with care.	Officeholders and titles are published exactly as the Foundation confirms, with an approval step; the platform takes no position.
Funding and accountability.	Donors and partners want to see where support goes.	Donation and pledge flow, programme progress bars and a transparency dashboard.

2.4 Who the platform serves

Audience	What they need	Where they meet the platform
Youth and students	Learn the language and history, find events, earn recognition	App, website
Elders and knowledge holders	Have their knowledge recorded and respected	Assisted contribution (a youth ambassador helps), admin review
Clan leaders	Maintain their clan's page and members, submit corrections	Clan self-service in the admin, website
Diaspora	Stay connected, learn, give, attend events	Website, app, diaspora corner
Researchers, media and schools	Citable, sourced material and open data	Website, library, open data API
Partners and donors	See programmes, progress and transparency	Website, programmes, transparency page
Foundation staff	Publish, review, report and communicate	Admin control panel

2.5 Guiding principles

1. **Community authority.** The Foundation decides what is true and what is published. The platform records, shows sources and enforces review. It never invents culture.
2. **Offline first.** We design for the weakest connection, not the best one.
3. **Plain language.** Clear words, large touch targets, readable type and simple flows, so that a first-time smartphone user can take part.
4. **Own your data.** Content, accounts and contributions belong to the Foundation and can be exported at any time.
5. **Built to last.** Boring, well-understood technology that local developers can maintain on affordable hosting, with documentation and no lock-in.
6. **Respect for knowledge.** Consent, attribution and restricted-access protocols are built in from the start (Chapter 7).

CHAPTER 3

Our Recommendation: One Platform, Three Surfaces

Why the website and the mobile app belong together, and how one shared platform gives the Foundation more reach for less cost.

3.1 The recommendation in one paragraph

Build a single platform with **one backend and three surfaces**: a public website for reach, credibility and search; an admin control panel where the Foundation's staff run everything; and the Lango Mobile App for daily use, offline learning and field contribution. Deliver it in a way that the Foundation owns, can run with its own people, and can extend for years without rebuilding.

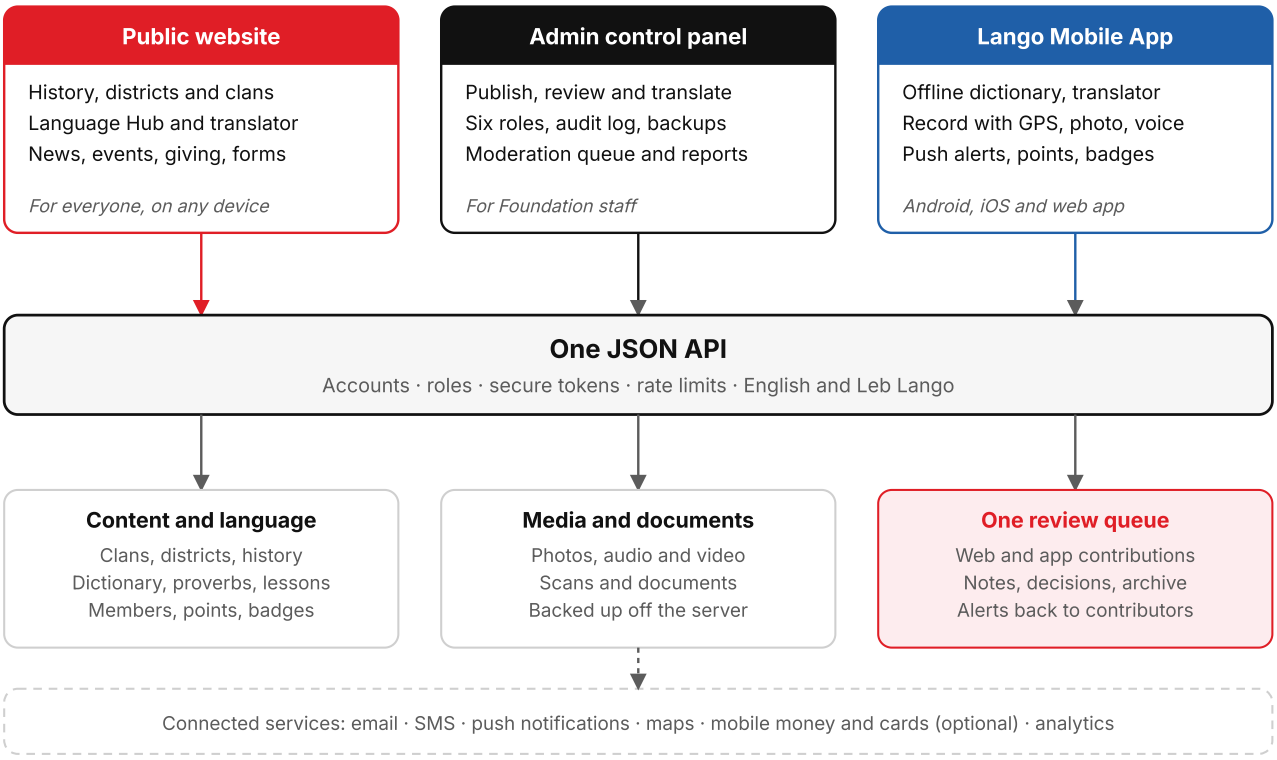


Figure 3.1: One platform, three surfaces. Every surface talks to the same JSON API, database and review queue.

3.2 Why not the website alone, or the app alone?

A website and an app are good at different things. Choosing only one leaves a gap that the Foundation's community will feel.

Need	Website	Mobile app
Be found on Google, shared in a link, cited by the media	**Strong.** Every page has its own address, search-engine markup and share preview	Weak. App content is not indexed in the same way
Read long histories and use large tables and maps	**Strong** on desktop and tablet	Good on a phone for short reading
Work with no internet	Limited (cached pages only)	**Strong.** Offline dictionary, queued contributions, saved tickets
Record a voice, a photo or a GPS point at the source	Possible, but clumsy in a browser	**Strong.** Native camera, microphone and location
Remind and re-engage people	Email only	**Strong.** Push notifications, word of the day
Serve people who do not install apps	**Strong.** Nothing to install	Via the installable web app (PWA)
Staff administration, reporting and moderation	**Strong.** A desktop workspace is the right tool	Not suitable
Reach older users and institutional partners on desktop	**Strong**	Not the primary channel
Reach youth and the diaspora on phones	Good	**Strong.** Installed, always at hand

The conclusion is simple. **The website is the Foundation's public face and its control room; the app is its daily companion and its field recorder.** Together they cover every audience in Chapter 2.

3.3 One backend: what "shared" means in practice

- **One set of accounts.** A member signs in once, on the website or in the app, with the same email or phone number and password.
- **One content store.** The history chapter an editor publishes in the admin panel is the same chapter the app shows, in English and in Leb Lango.
- **One review queue.** Every contribution, word suggestion, proverb or directory profile, whichever surface it came from, lands in the same moderation workspace, with the same status model and the same notifications to the contributor.
- **One API.** The app uses the same documented interface that powers the open data page. Any future channel (a kiosk, an SMS service, a partner's portal) can use it too.
- **One place to fix things.** A correction made once is corrected everywhere.

3.4 Who uses which surface

Audience	Website	Admin panel	Mobile app
Visitors, students, researchers, media	●		○
Members and contributors	●		●
Diaspora	●		●
Clan leaders	●	● (own clan only)	○
Moderators and language experts	○	●	
Editors and communications staff	○	●	
Administrators and leadership	○	●	
Partners and donors	●		○

● primary surface · ○ occasional use

3.5 The three packages

To give the Foundation a clear decision, we propose three packages. Prices are in Chapter 13; the scope is summarised here.

Scope	Option A: Website and admin	Option B: Complete platform *(recommended)*	Option C: Complete platform plus growth year
Public website, bilingual	●	●	●
Admin control panel and roles	●	●	●
Language Hub (dictionary, translator, proverbs, lessons)	●	●	●
Community contribution engine on the website	●	●	●
Lango Mobile App (Android, iOS, web app)		●	●
Offline dictionary and offline contributions		●	●
Push notifications and member notifications		●	●
Store publishing (Google Play, Apple App Store)		●	●
Training and handover	●	●	●
90 days of hypercare	●	●	●
Content digitisation support (supervised interns)			●
12 months of support and quarterly improvements			●

Scope	Option A: Website and admin	Option B: Complete platform *(recommended)*	Option C: Complete platform plus growth year
Launch campaign and digital engagement programme			•

Option A is complete and valuable on its own, and every part of it is reused if the Foundation adds the app later. We recommend Option B because the incremental cost of the app is much lower when it is built alongside the platform than when it is added to a finished one, and because the app is where the community's contributions will come from.

3.6 Starting from a working demonstrator

The delivery plan in Chapter 11 begins with the demonstrator, not a blank page:

- **Risk is lower.** The hardest engineering problems (offline sync that cannot create duplicates, the moderation workflow, bilingual content, push notifications) are already solved and tested.
- **The Foundation can see before it signs.** Appendix B shows the demonstrator, and we can give the Foundation a live walk-through at the discovery workshop.
- **Time goes where it matters.** The first weeks of the project are spent on the Foundation's identity, structure and content, not on infrastructure.
- **Nothing is hidden.** The demonstrator shows exactly what is built and what is still to do (the roadmap items in Section 6.9), so expectations are clear from day one.

> **A note on the name.** The demonstrator was built under the working title *Lango Cultural Institution*. In this project every screen, logo, colour, text and app store listing will carry the Foundation's own name and identity.

CHAPTER 4

The Public Website

A fast, bilingual, brand-led home for the Foundation's history, language, clans, heritage and community, designed to be read on a phone first.

4.1 Experience and design

The website will carry the Foundation's own identity: its logo, colours, photography and tone. The demonstrator shows the approach: a contained editorial layout, a clear top bar with contact details, language switch and sign-in, and a full navigation with expandable sections on phones.

Design principle	What it means for visitors
Mobile first	Pages are designed for a phone screen and then enlarged, because most visitors will arrive on a phone
Bilingual	A one-tap English / Leb Lango switch on every page; content and interface can each be translated by the Foundation's experts
Brand-led	The Foundation's logo, colours and photography, with the demonstrator's clean, editorial look as the starting point
Readable	Generous type size and contrast, WCAG 2.1 AA as the target, keyboard-friendly navigation
Light	Optimised images, sensible page weight and a lightweight mode for slow connections
Findable	Search-engine friendly addresses, structured data, a sitemap and share previews on every page

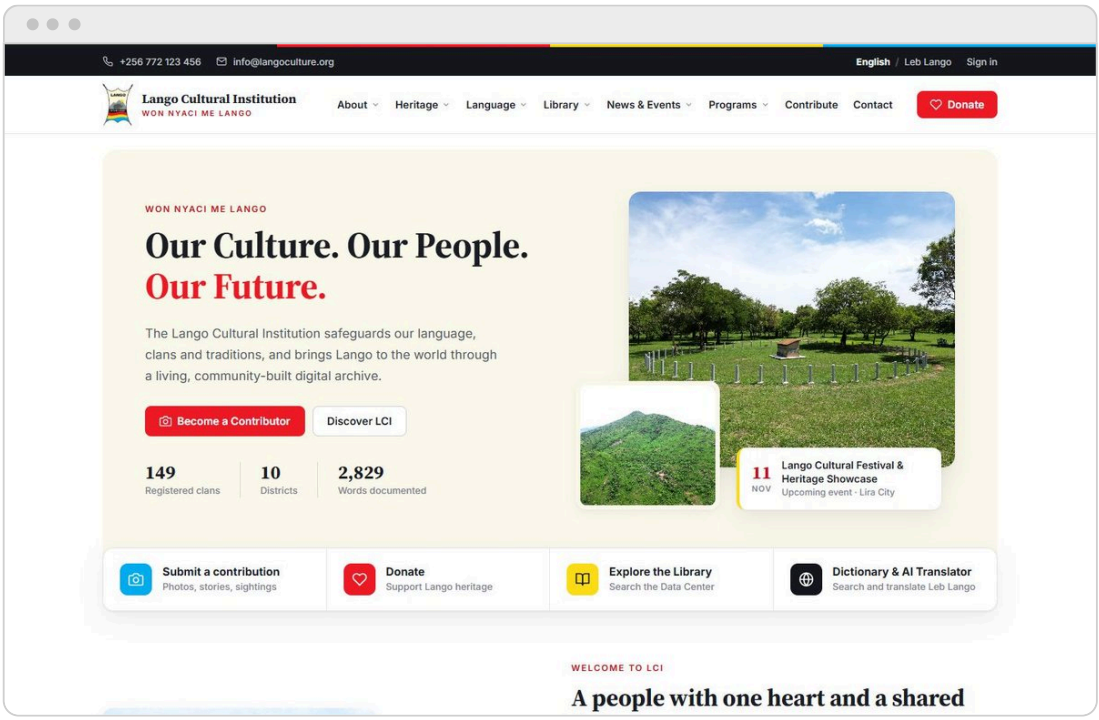


Figure 4.1: The demonstrator home page: hero, quick actions, word of the day, clans, districts and news.

4.2 Site map and features

Section	Pages and features	What visitors can do
Home	Hero, quick actions, word of the day, featured notable Lango, news, events, programmes, giving call	Orient themselves and reach any section in one tap
About and Leadership	Mission and vision, the Won Nyaci, the cabinet and elders, organisation details	Learn who leads the Foundation and how it works
History	Ten chapters with a table of contents, a timeline, a glossary of terms and a list of sources	Read the story of the Lango people with every claim sourced
Districts and Lira City	An atlas of the nine districts and Lira City, a comparison table, a map, and a page per district (demography, economy, places, history)	Compare and explore the sub-region
Clans	A searchable Clan Explorer with a page per clan showing totem, district, summary and clan leader	Find a clan, learn about it, correct or add information
Notable Langos	Biographies of notable people by category, with portraits where licensed photographs are available	Discover the people who shaped Lango
Heritage	Ceremonies, attire and crafts, music and dance, symbols, and heritage sites on a map	Explore living culture in detail
Biodiversity	A register of species with a community sightings map	Record and browse local wildlife and plants
Language Hub	Dictionary, English to Leb Lango translator, proverbs, greetings, lessons, "suggest a word" (Section 4.3)	Learn, look up and contribute
Library and Media Centre	Documents with downloads; photo, video, audio and podcast galleries	Find sourced documents and listen to recordings
News and Blog	Categories, bilingual posts, comments with moderation, share buttons	Follow Foundation news
Events	Calendar, event pages, online registration, printable ticket with code	Register and attend
Programmes and Transparency	Programmes with progress and a sponsor button; confirmed donation totals	See what is funded and how it is going
Get Involved	Contribute, scholarship, grant, volunteer, partner, clan registration, diaspora forms	Take part, with the clan picker on every form
Support	Donation and pledge flow, with live payment as an optional upgrade	Give securely and receive confirmation
Professionals Directory	Verified profiles of Lango professionals	Find and connect with skills
Cultural Assistant	A question box that answers only from reviewed content, with sources	Ask a question and see where the answer comes from
Data Center and Search	Faceted search across the whole site	Find anything in seconds
Open Data and Developers	Documented JSON endpoints and an open data licence page	Reuse data for research and schools
360 degree tours	Interactive panoramic viewer for sites and events	Visit places remotely

Section	Pages and features	What visitors can do
Legal	Privacy and Terms pages	Understand how their data is handled

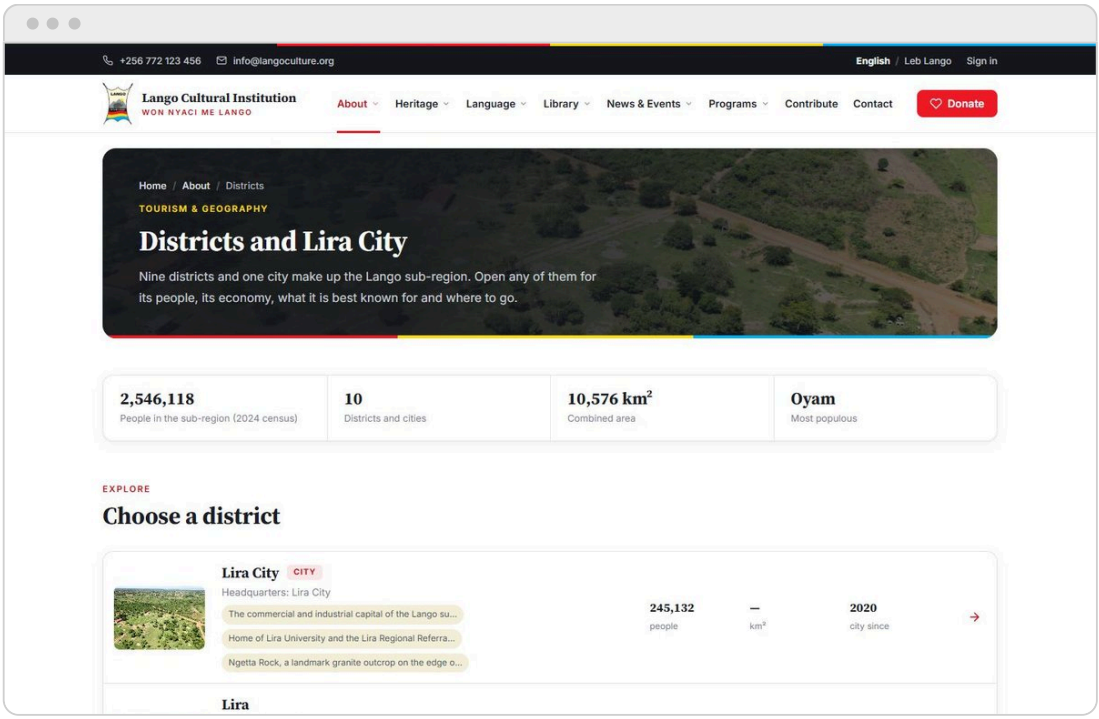


Figure 4.2: The District Atlas, with a comparison table and a map of the sub-region.

4.3 The Language Hub

Language is the heart of the platform, and the demonstrator already shows how it should work.

- **Dictionary.** Search in either direction, filter by topic or letter, and browse A to Z. The search folds special characters, so typing *e* finds *ɛ* and *ng* finds *ŋ* on a plain keyboard.
- **Translator.** Translates between English and Leb Lango using the dictionary, not machine translation, and says so. Every published dictionary entry is available immediately.
- **Proverbs and greetings.** Published only when sourced or approved; none are pre-loaded unless the Foundation supplies them.
- **Lessons and phrasebook.** Short structured lessons and everyday phrases for learners.
- **Suggest a word or proverb.** Anyone can propose an entry. A duplicate check runs first; a language expert edits and approves; the entry then appears with a **Community** label until verified.
- **Verification labels.** Each entry shows whether it is *Verified by experts*, *Community* or *Unverified*, so users always know how much to trust it. The demonstrator already holds a starter dictionary of 2,829 entries, every one marked *unverified* until the Foundation's language experts review it.

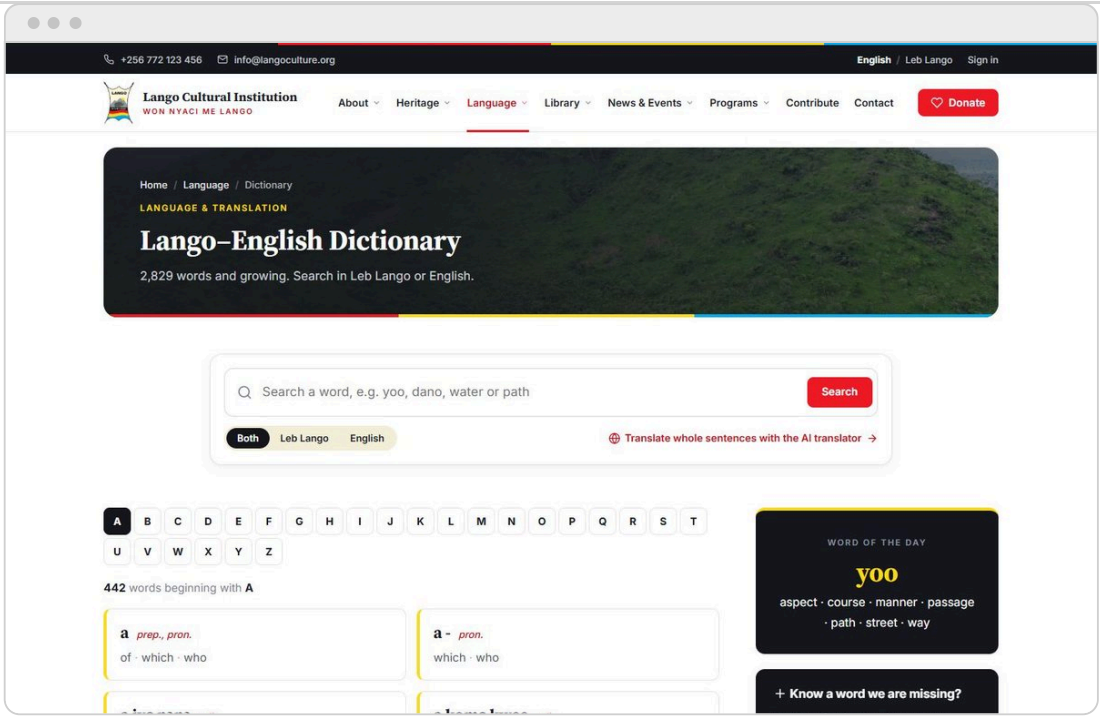


Figure 4.3: The dictionary with topic filters, A to Z browsing and a verification status on every entry.

4.4 History and heritage, done carefully

- **Sourced chapters.** History chapters are compiled from published sources, shown on the page, and flagged for the Foundation's historians to review before launch.
- **Timeline and glossary.** Key dates and terms are structured data, so they can be reused across the site and the app.
- **Heritage items** carry category, description, photographs, audio and, where relevant, a map location.
- **Corrections welcome.** Every page will offer a route for users to suggest a correction, which enters the same review queue.

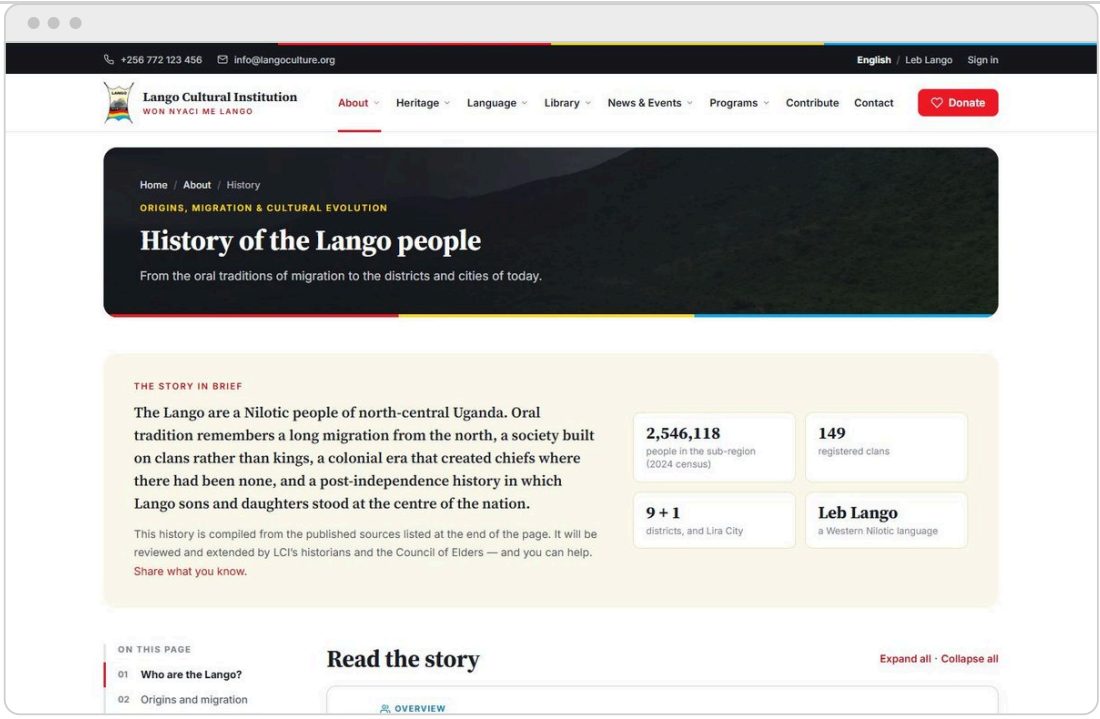


Figure 4.4: History, with a table of contents, accordion chapters and a sources list.

4.5 Community participation

The website lets people do more than read.

- **One simple contribution form.** A visitor can submit a cultural observation, a natural heritage sighting, a word, a proverb, an oral history, a document or a heritage site. They can create an account on the spot, choose their clan, add GPS and upload photos, audio and files. They receive a **tracking code** to follow the submission.
- **Accounts, points, levels and badges** recognise contributors; a leaderboard celebrates the most active contributors.
- **Comments** on news and posts are moderated before they appear.
- **Clan picker** on every registration (account, contribution, scholarship, grant, volunteer, diaspora, event ticket, directory profile) with a "my clan is not listed" option that alerts administrators and builds the register over time.

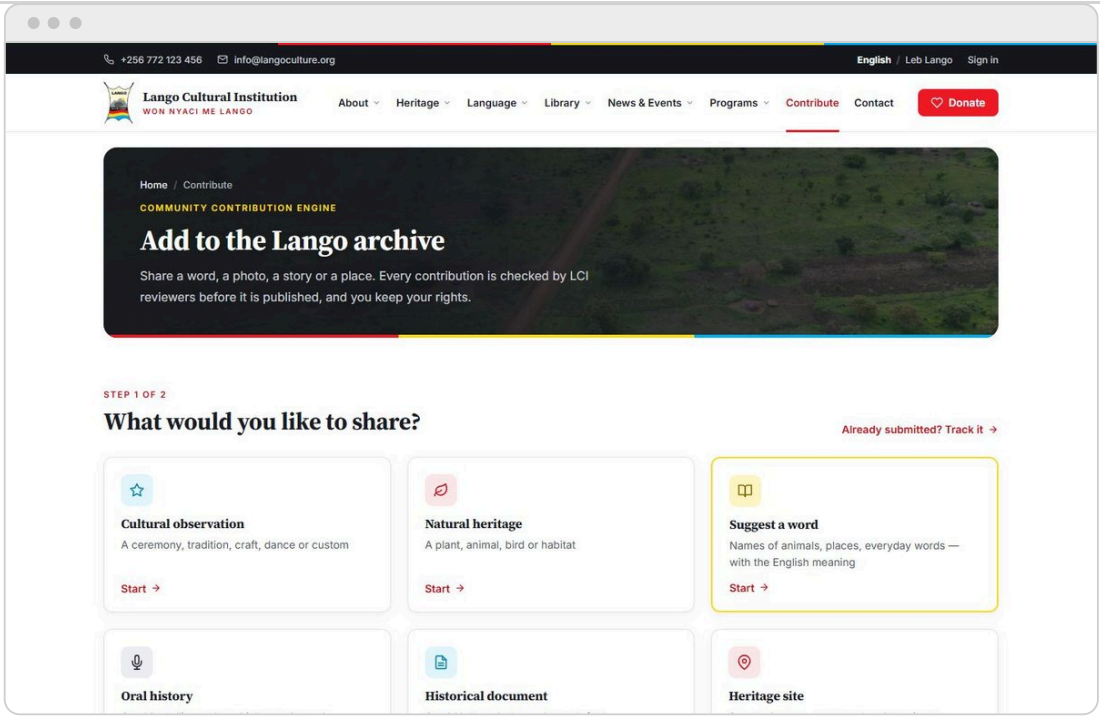


Figure 4.5: The community contribution form with seven contribution types, optional account creation and a tracking code.

4.6 Diaspora corner

A dedicated diaspora section explains how Langos abroad can connect, give, volunteer and contribute, with a registration form that records country and clan. It gives the Foundation a measurable channel to an audience that is otherwise hard to reach.

4.7 Discoverability, analytics and sharing

- Search-engine friendly addresses, titles and descriptions on every page, an XML sitemap, and share previews (Open Graph) for WhatsApp, Facebook and X. Structured data for events and articles will be added in this engagement.
- Privacy-conscious analytics, added in this engagement, that show the Foundation which content people use and where they come from, without tracking individuals.
- A newsletter subscription with an exportable subscriber list.

4.8 Quality targets

Area	Target (to be confirmed at discovery)
Accessibility	WCAG 2.1 AA for contrast, keyboard navigation and labels
Performance	Good Core Web Vitals on a mid-range phone on a 4G connection
Browser support	Current versions of Chrome, Edge, Firefox and Safari, on desktop and mobile
Availability	[TBC: target uptime, for example 99.5 percent, per Chapter 14]
Languages	English and Leb Lango for interface and for all content the Foundation chooses to translate

CHAPTER 5

The Admin Control Panel

A role-based workspace where the Foundation's own people publish, review, translate and report, with every change recorded and reversible.

5.1 Designed for the people who will use it

The admin panel is where the platform stays alive after launch, so we design it for the Foundation's staff and volunteers rather than for developers. The demonstrator already shows the approach: one consistent layout, a live notification bell, a dashboard that opens on what needs attention, and every managed content type edited through the same simple forms.

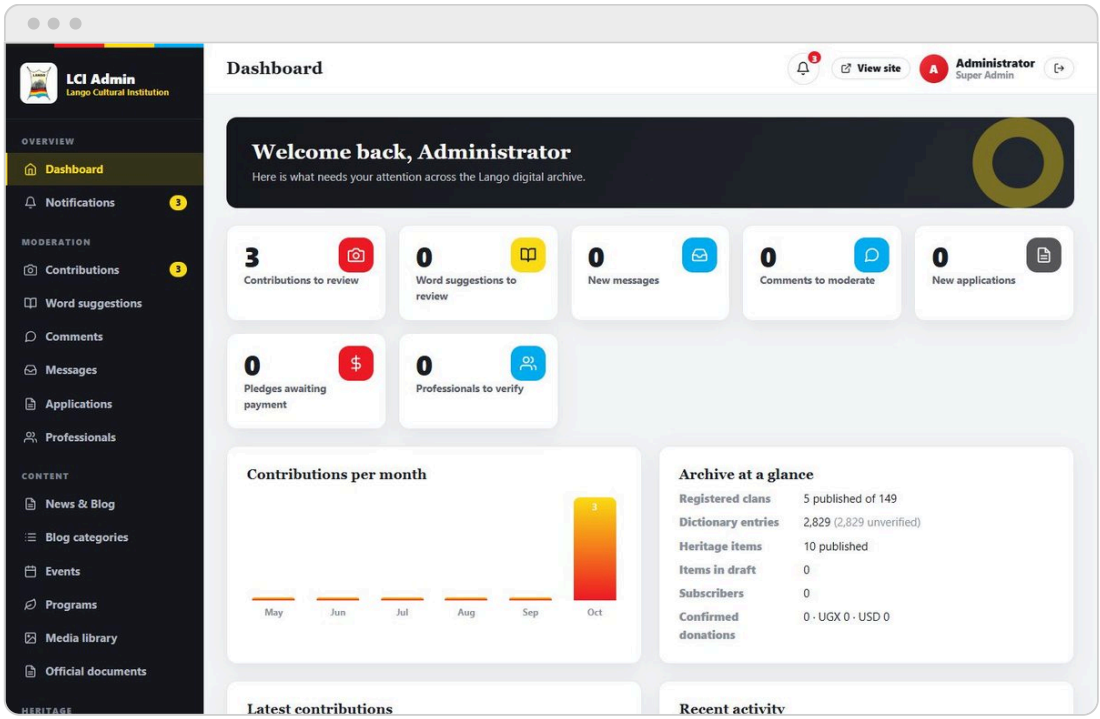


Figure 5.1: The admin dashboard: items waiting for review, contributions per month, the archive at a glance, latest contributions and quick actions.

Principle	In practice
No code for daily work	Pages, news, events, history, districts, clans, dictionary entries and more are edited in forms with rich-text editors and image uploads
Everything is reversible	Version history with restore on content; an audit log of who did what and when
Right access for each person	Six roles, so a clan leader sees only their clan and a moderator sees only the review queue
Never miss anything	A live bell and a Notifications page for every new submission, comment, message, application, donation, directory profile, event registration, new member and "clan not listed" alert
Built to extend	A new content type is added by registering one entry, which keeps the platform cheap to grow

5.2 Roles and permissions

Role	Typical person	Can do
Super admin	Platform owner, Kakebe during support	Everything, including users, roles, settings, backups
Admin	Foundation operations lead	Everything except platform-level settings
Editor	Communications and content staff	Create and publish news, pages, history, heritage, events, programmes, media
Moderator	Language experts, community managers	Review contributions, word suggestions, comments and messages
Clan leader	Verified leader of one clan	Edit their own clan's page and members; changes need review, so they cannot self-publish
Contributor	Any registered member	No admin access; submits content from the website or the app

The final role set, and which permissions each role holds, is agreed in discovery. Custom roles can be added.

5.3 What can be managed

The demonstrator manages **30 content types** through a single, consistent engine, plus bespoke screens for the tasks that need more. They fall into six groups.

Group	Managed items
Culture and language	Dictionary, proverbs, lessons, clans (with bulk add and CSV import), districts, notable Langos, history chapters, timeline, glossary, heritage items, species, 360 degree tours, leaders
Content and media	News posts and categories, events, programmes, media (photo, video, audio, podcast), documents, partners
Community and moderation	Contributions, word and proverb suggestions, comments, messages, applications, professionals, subscribers
Finance	Donations (confirm or fail, CSV export), programme progress, transparency figures
Assistant	Knowledge base entries, chat log, flagged answers for review
System	Users and roles, site settings and branding, translations, notifications, audit log, database backup, data export

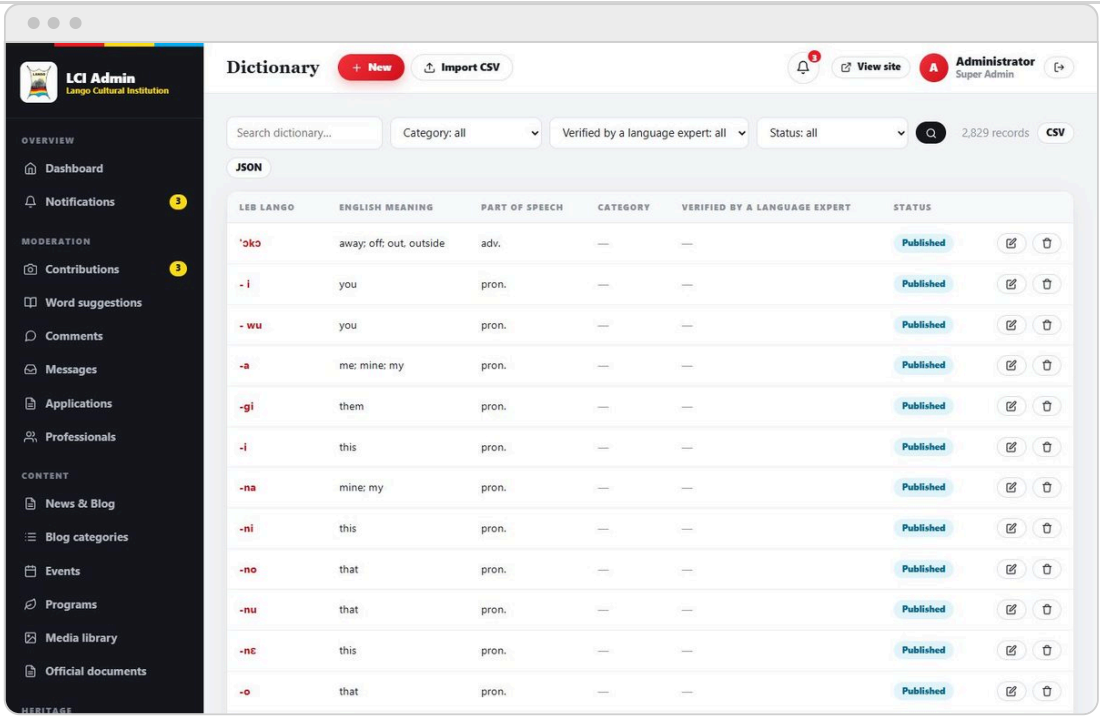


Figure 5.2: Managing the dictionary: filters, status, verification and bulk import.

5.4 The moderation workspace

This is the screen that protects the Foundation's credibility. Every contribution arrives with its type, description, location, media, clan and contributor, and the reviewer can:

1. **Open and read** it with its photographs, audio and map pin.
2. **Add a review note** that the contributor sees.
3. **Edit** the text, for example to correct spelling.
4. **Approve, publish, return for changes or reject.** Each action notifies the contributor in the app and on the website, and sends a push notification if allowed.
5. **Award points** for good contributions.
6. **Add to the archive**, which turns an approved contribution into a permanent record (a dictionary entry, a heritage site, a document and so on).

Chapter 7 describes the full lifecycle and the review tiers.

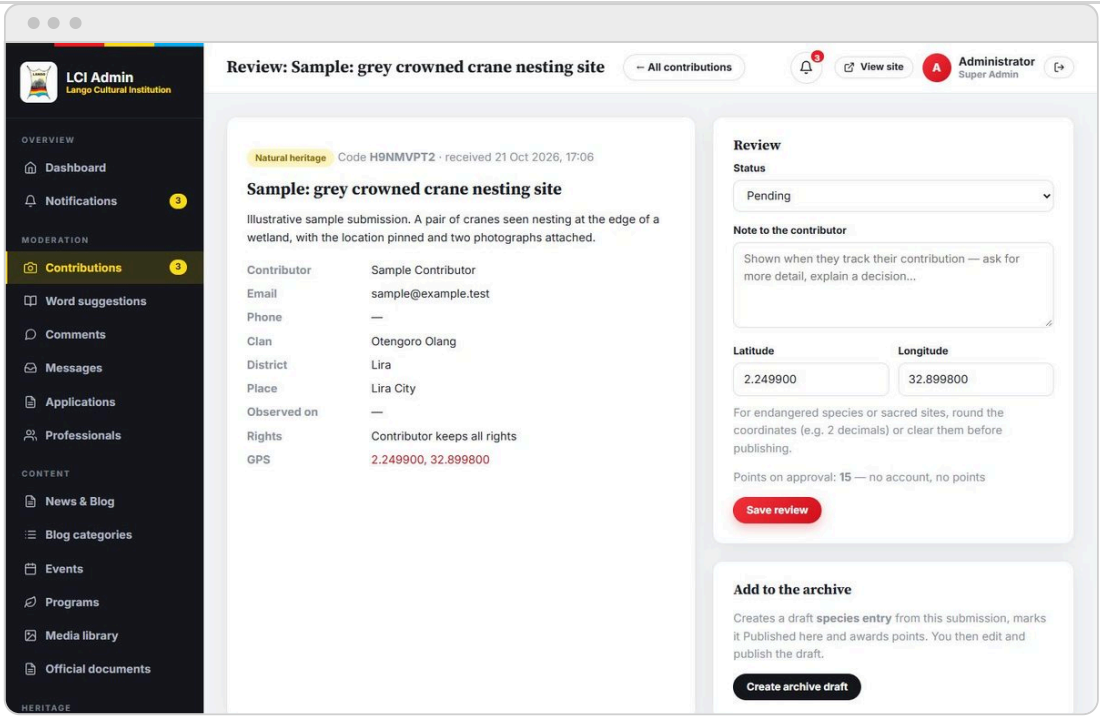


Figure 5.3: Reviewing a contribution with its media, location, note field and status actions.

5.5 Translations and the Leb Lango interface

The **Translations** screen lists every phrase of the website and the app interface. On first run, the app registers its own phrases so they appear ready to translate. The Foundation's language experts enter the Leb Lango wording; anything not yet translated shows in English. **Nothing is machine-translated or invented.** Content fields such as news and history also have a Leb Lango version beside the English one.

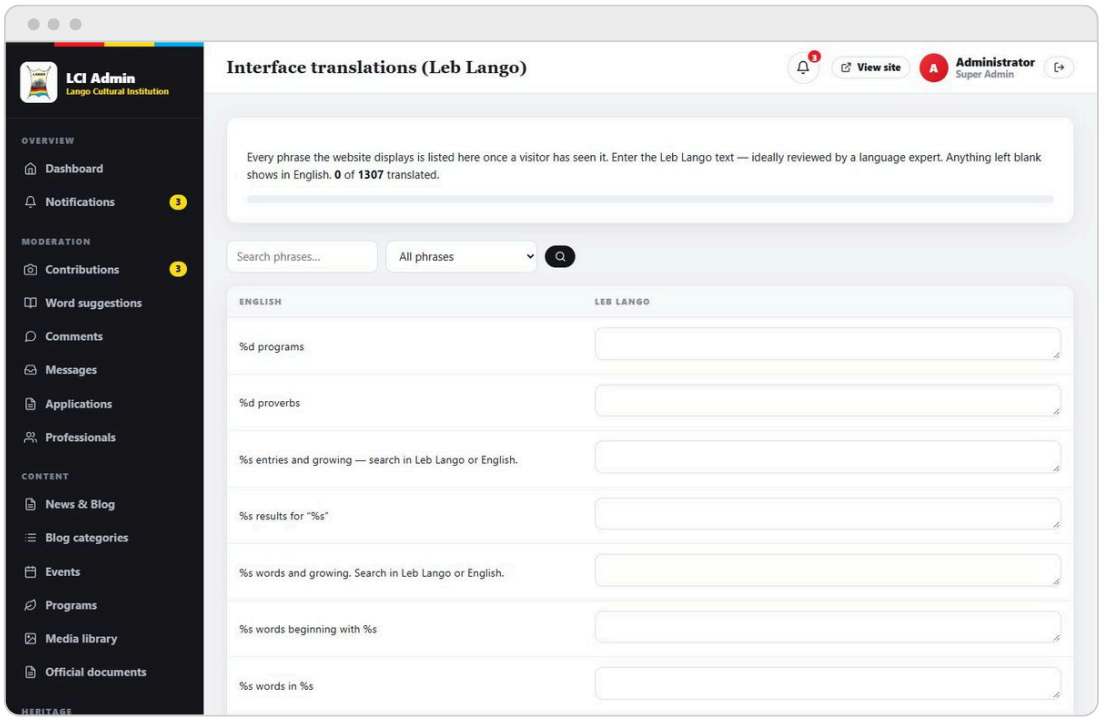


Figure 5.4: Translations: each English phrase beside the field for its Leb Lango wording.

5.6 Reporting, export and safety

- **Dashboards** with counts of contributions, words, members, registrations, donations and applications.
- **Exports** to CSV and JSON for the main lists, and a full **database backup** (SQL) on demand.
- **Audit log** of administrative actions with the user, time and item.
- **Version history and restore** on content, so a mistaken edit is never permanent.
- **Forced password change** at first sign-in; login throttling and rate limits on sensitive actions.

5.7 What the Foundation controls without Kakebe

After handover, Foundation staff can independently: publish and edit any content; translate the interface; review and approve contributions; add and import clans, dictionary entries and proverbs; manage events, tickets and registrations; confirm donations; change the logo, photographs and contact details; create users and assign roles; export data; and download backups. Kakebe is needed for new features, security updates and infrastructure, which is exactly what the support plans in Chapter 14 cover.

CHAPTER 6

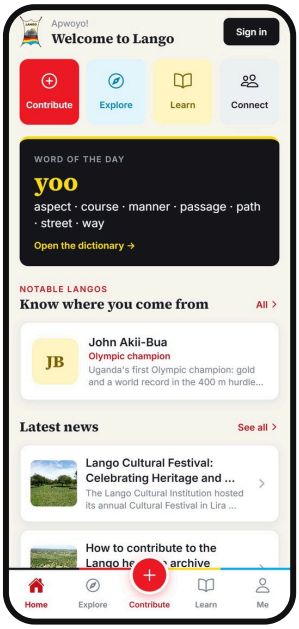
The Lango Mobile App

The whole platform in a pocket: offline dictionary, guided contribution from the field, notifications and recognition, on Android, iOS and the web.

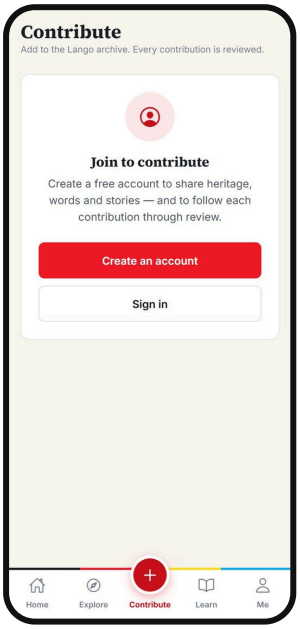
6.1 Why an app

For most people in Lango the phone is the computer. An installed app is one tap away, works when the signal drops, and can use the things only a phone has: the camera, the microphone, GPS and push notifications. That makes it the right tool for two jobs the website cannot do well:

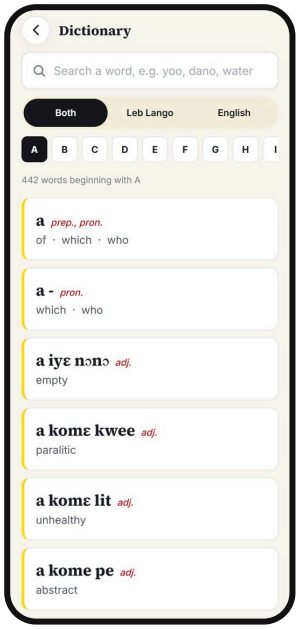
1. **Daily learning and connection.** A word of the day on the home screen, an offline dictionary, a phrasebook and a translator that works without data.
2. **Capturing knowledge where it lives.** An elder telling a story, a heritage site in the bush, a ceremony in a village: record the audio, take the photo, drop the GPS pin, and let the app send it when the connection returns.



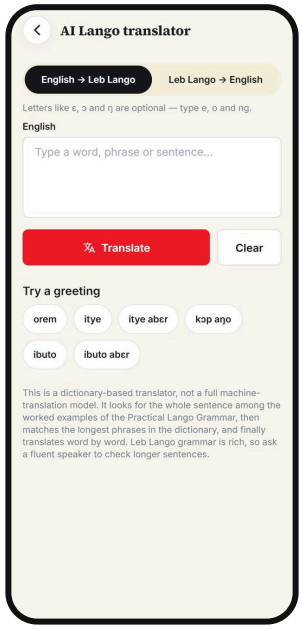
Home: greeting, word of the day, points and level, news and events.



Contribute: seven guided forms with GPS, camera, audio and files.



Dictionary: works offline once downloaded.

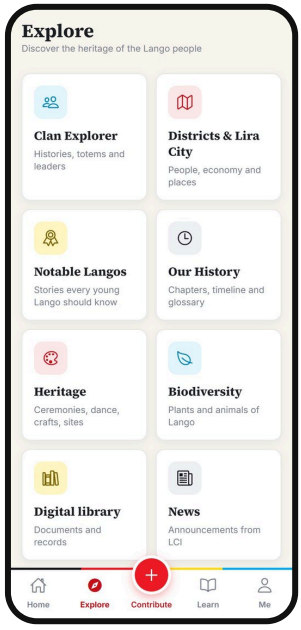


Translator: English and Leb Lango, with no machine translation.

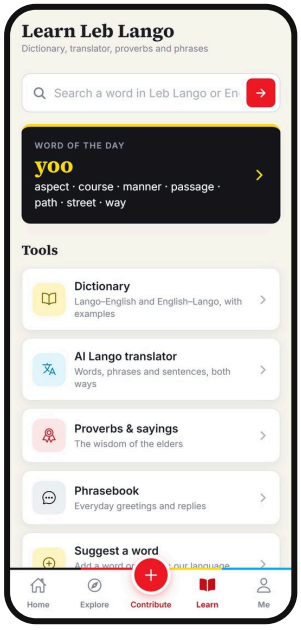
6.2 What the app does

The app is organised into five tabs, with one codebase serving Android, iOS and the installable web app.

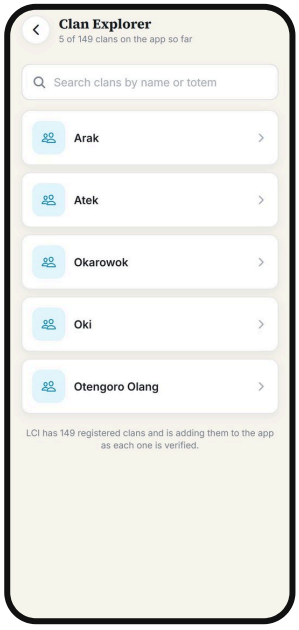
Area	Features
Onboarding and account	Language choice (English or Leb Lango); sign up with clan; sign in with email or phone; browse as a guest; delete the account at any time
Home	Greeting, quick actions, word of the day, the member's points and level, a notable Lango, news, events, giving call
Explore	Clans, districts and Lira City, notable Langos, history and timeline, heritage, biodiversity, digital library
Contribute	Seven guided forms with GPS, camera, gallery, audio recording and file upload; tracking code; works offline (Section 6.3)
Learn	Dictionary (offline once downloaded), English to Leb Lango translator, proverbs, phrasebook, suggest-a-word
Connect	Events with ticket registration, professionals directory, the source-bound Cultural Assistant, news, leaderboard
Support	Donation pledges, programmes and transparency figures
Me	Points, Bronze to Platinum levels, badges, my contributions and their status, notifications (in-app and push), settings



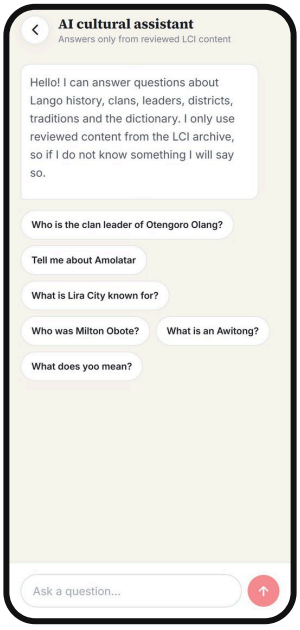
Explore: every section of the archive from one screen.



Learn: language tools in one place.



Clans: search and browse, with the clan leader shown.



Cultural Assistant: answers only from reviewed content, with sources.

6.3 Built to work with no internet

Connectivity in northern Uganda is uneven and data is expensive, so the app is designed around the worst connection, not the best.

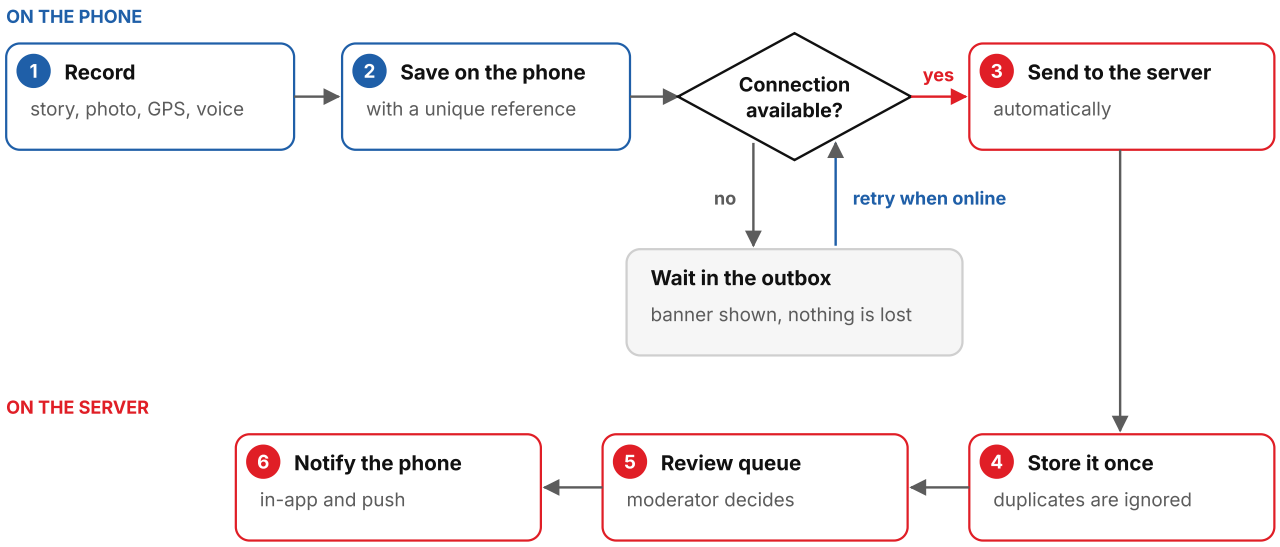


Figure 6.1: Offline-first behaviour. A contribution is saved on the phone, queued, and sent when the connection returns, with a unique reference so it can never be stored twice.

Behaviour	How it works
Stale-while-revalidate	Every list and page shows the last saved copy immediately, refreshes in the background, and keeps showing the saved copy if there is no internet
Offline dictionary	One tap downloads all published words. The download is versioned, so an update downloads nothing when nothing has changed. Search works in both directions with the same plain-keyboard folding as the website
Offline contributions	If a send fails, the contribution and its photo, audio or file are copied into the app's own storage and wait in a queue, with a banner. They are sent automatically when the connection returns
No duplicates, ever	Each contribution carries a unique reference, so a retry after a dropped connection cannot create a second copy on the server
Tickets and tracking codes	Saved on the phone so they open offline at the event gate or in a conversation with the Foundation
Secure sign-in	A token kept in the phone's secure store (Keychain on iOS, Keystore on Android); a rejected token signs the person out cleanly

6.4 Contributing from the field

The app lets members contribute seven kinds of knowledge with the right fields for each:

Contribution type	Typical use
Cultural observation	A ceremony, a craft, a custom, a place
Natural heritage	A plant, animal or habitat, with photo and GPS
Word	A Leb Lango word, with meaning and example
Proverb	A proverb with its meaning and context
Oral history	A story or account, with audio and the name of the speaker (with consent)
Document	A scan or photograph of a letter, certificate or record
Heritage site	A place with a GPS pin, photographs and description

Every contribution shows its **status** (pending, under review, returned, approved, published) to the contributor, with the reviewer's note. When a reviewer acts, the contributor receives an in-app notification and, if they allowed it, a push notification. This feedback loop is what keeps contributors coming back.

6.5 Recognition without pressure

Members earn **points** for approved contributions and move through **Bronze, Silver, Gold and Platinum levels**, earning **badges** and appearing on a **leaderboard** of top contributors. The aim is recognition for people who preserve the culture, not competition for its own sake. The Foundation decides at discovery which recognition features are shown publicly, and how contributors can use a display name instead of their own.

6.6 Notifications and re-engagement

- **In-app notifications** for every contribution decision, with a read/unread state.
- **Push notifications** through Firebase Cloud Messaging on Android and Apple Push on iOS, sent when a reviewer acts and, in later phases, for events, news and the word of the day.
- **Event registration** with a ticket and code stored on the phone.
- **Word of the day** on the home screen to build a daily habit.

6.7 Languages and accessibility

The interface is available in **English and Leb Lango**. Each phrase is looked up in the translations the Foundation's experts enter in the admin panel; anything not yet translated shows in English. Nothing is machine-translated. The app uses large touch targets, readable type (Inter and Source Serif 4) and clear states for loading, errors and empty lists, so a first-time smartphone user is never left on a blank screen.

6.8 Platforms and distribution

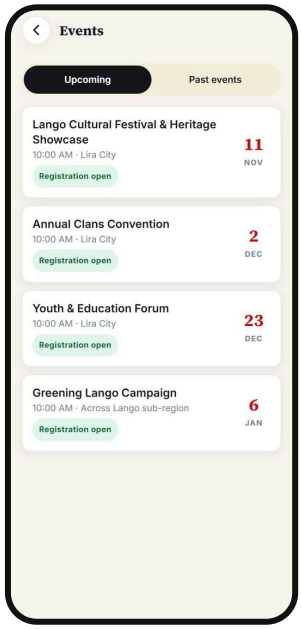
Channel	How people get it	Notes
Android	Google Play, or a direct APK link for pilots	Built in the cloud; the Foundation's own Play Console account is recommended
iOS	Apple App Store	Requires an Apple Developer account in the Foundation's name
Installable web app (PWA)	Open a link in the browser and choose *Add to Home screen*	No store needed; ideal for people on low-storage phones and for instant sharing

The app is built with **Expo and React Native**, so one codebase produces all three. Builds run in the cloud, so no special equipment is needed. Over-the-air updates let us fix small problems without a store review. We register and publish the app in the Foundation's own store accounts so that ownership stays with the Foundation.

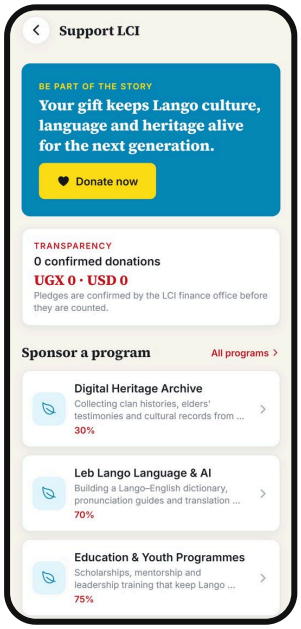
6.9 Roadmap items beyond the first release

These are valuable but not required for a successful launch. We list them so the Foundation can plan and prioritise.

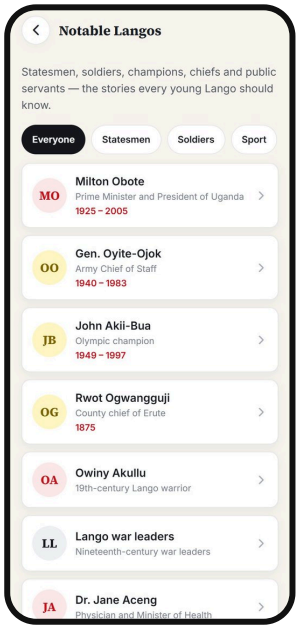
Item	Why it matters
SMS and USSD fallback	Reaches people on basic phones and in areas without data
One-time-code (OTP) sign-in	Easier sign-in for first-time and low-literacy users
Community forum and clan discussion groups	A moderated space for clans and the diaspora
Maps and 360 degree tours inside the app	Brings sites and events to life
Live mobile-money and card payments	MTN MoMo, Airtel Money and cards for giving, tickets and sponsorships (Kakebe already runs these in other products)
Extended AI assistant	A richer assistant that still answers only from reviewed content



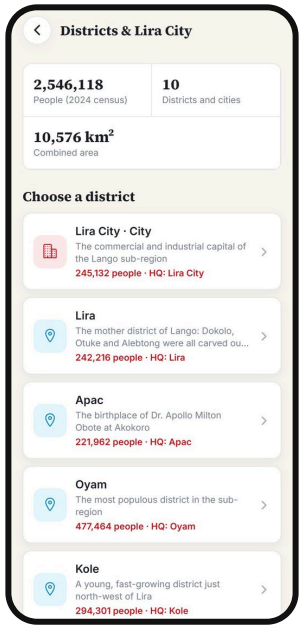
Events: registration with ticket and code.



Support: giving, programmes and transparency.



People: notable Langos.



Districts: the sub-region at a glance.

CHAPTER 7

Community Contribution and Content Governance

How knowledge enters the archive, who decides what is true, and how the platform protects the Foundation's credibility and its people.

7.1 The principle: the community decides, the platform enforces

A digital archive is only as trusted as its weakest entry. The Foundation's reputation therefore depends on one rule that is built into the platform and into this proposal:

> **The platform never invents culture.** Clans, proverbs, translations, biographies and history come only from the Foundation, from published sources that are shown on the page, or from community contributions that an authorised reviewer has approved. Where something is missing, the platform shows it as empty, not made up.

Everything in this chapter follows from that rule.

7.2 The contribution lifecycle

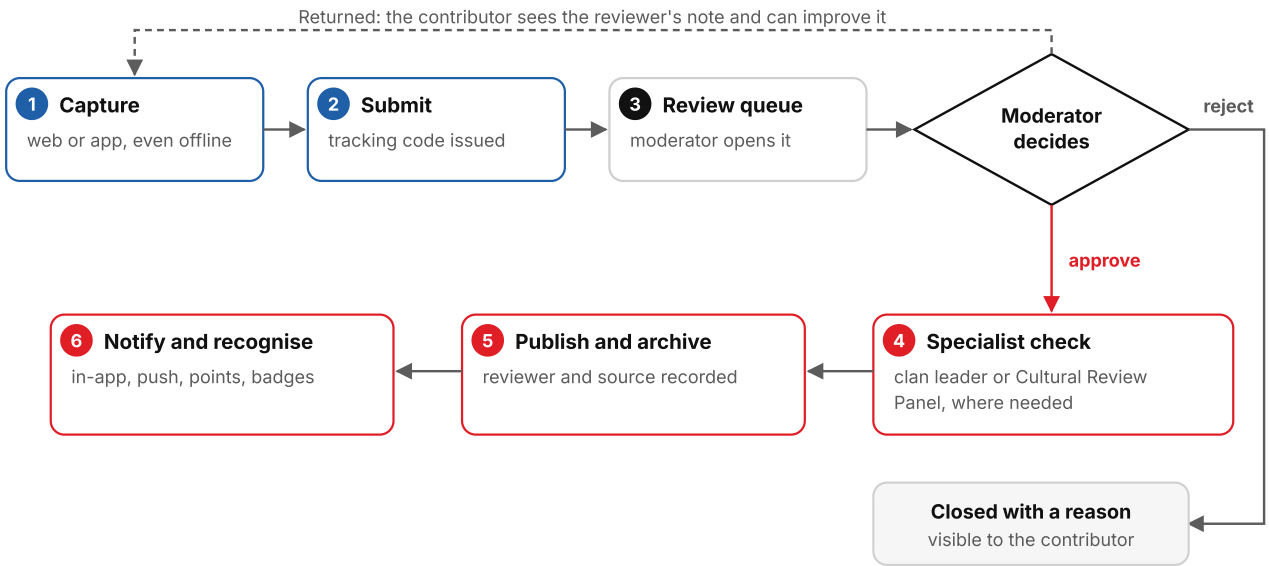


Figure 7.1: The lifecycle of a contribution, from capture on the phone or website to a published, attributed archive record.

Step	What happens	Who acts
1. Capture	The contributor fills one of seven guided forms on the website or the app, adding location, photos, audio or files; offline if needed	Contributor
2. Submit	The submission receives a **tracking code** and a status of *Pending*. Duplicates are prevented, and rate limits stop floods	System
3. Alert	A bell notification and a new row in the review queue tell the moderators something is waiting	System
4. Review	A moderator reads it, adds a note, corrects spelling, and decides: approve, return for changes or reject	Moderator
5. Specialist check	Where the content needs it (a clan's history, a sacred practice, a contested fact), it is routed to the clan leader or the Foundation's elders' panel before publication	Clan leader, elders' panel
6. Publish and archive	The record is published or added to the archive as a dictionary entry, heritage site, document and so on, with its contributor and reviewer recorded	Moderator, editor
7. Notify and recognise	The contributor is notified in-app and by push, and receives points and badges	System

7.3 Status model

Status	Meaning	Visible to the contributor
Pending	Received and waiting for review	Yes, with the tracking code
Under review	A reviewer has opened it	Yes
Returned	The reviewer needs more information or corrections; the note explains what	Yes, with the note
Approved	Accepted by the reviewer	Yes
Published	Live on the website and the app	Yes
Rejected	Not accepted, with a reason	Yes

Anyone can check a submission's status with its tracking code, without logging in.

7.4 Review tiers and roles

The demonstrator already separates the roles that matter most:

- **Moderators** (language experts and community managers) review contributions, word suggestions and comments.
- **Clan leaders** can edit only their own clan and **cannot publish their own changes**. Every change they make is reviewed, so the clan's voice is respected without bypassing quality control.
- **Editors** publish Foundation content such as news, events and programmes.
- **Admins** manage users, roles and settings.

For this project we recommend adding a **Cultural Review Panel**, a small group of elders and historians nominated by the Foundation, who act as the final reviewer for sensitive content. [TBC: the Foundation to confirm panel composition and the categories that require panel sign-off.]

7.5 Trust labels and sources

Every record tells the reader how far to trust it.

Label	Meaning
Verified	Checked and approved by the Foundation's experts
Community	Contributed by a member and approved by a moderator, not yet verified by the panel
Unverified	Loaded from an external or starter source and awaiting review (for example the starter dictionary)
Source cited	History, districts and people show the published sources they come from

7.6 Cultural protocols: consent, attribution and restricted knowledge

Some knowledge is not meant for everyone, and some belongs to a family, a clan or a ceremony. We propose that the platform support the following, to be agreed with the Foundation at discovery.

Protocol	How the platform supports it
Informed consent	Recording and photography forms record that the speaker or owner has agreed to the use of their words and image
Attribution	Contributors and speakers are credited as they wish: by name, by clan, or anonymously
Restricted knowledge	A record can be marked *restricted* so that it is held in the archive but shown only to authorised roles, or not shown at all
Right of correction and removal	Every page will offer a route to ask for a correction or withdrawal, handled in the review queue
Children's content	Clear rules on photographs and recordings of children, with guardian consent
Neutral on contested matters	Leadership titles and officeholders are published exactly as the Foundation confirms, with an approval step; the platform takes no position on disputes

7.7 Ownership, licensing and attribution

- **The Foundation owns the platform's content and data.** Contributors keep authorship of what they submit and grant the Foundation a licence to publish and preserve it, which the contribution form states plainly.
- **Public reuse is a choice.** The Foundation decides which data is released as open data, under which licence, on the open data page. The default is to share nothing without approval.
- **Sources are respected.** Material from published sources is cited, and permission is sought for photographs, recordings and documents before they are published.

7.8 Keeping out spam and abuse

- **Rate limits** on sign-in, registration, uploads, the translator and the assistant.
- **Duplicate checks** on word suggestions and idempotent uploads.
- **Moderation before publication** for contributions, comments and directory profiles.
- **Account controls:** administrators can deactivate abusive accounts, and members can delete their own.

- **Report and correction routes:** a visible way on public pages to report content or request a correction, feeding the moderation queue (added in this engagement where not already present).

7.9 The Cultural Assistant

The assistant answers visitors' questions, for example *Who is the leader of this clan?* or *What is known about this district?*, **only from the register and the reviewed archive**, and it shows its sources. If it cannot answer, it says so and logs the question as a flagged item so that the Foundation can see what people are asking and fill the gap. It is deliberately not an open-ended chatbot: it cannot make things up because it has nothing to draw on except approved content.

CHAPTER 8

Architecture, Security and Data Protection

Plain, proven technology on affordable hosting, with security and privacy designed in and a clear route to grow.

8.1 Architecture at a glance

The platform is a classic, robust three-layer design. Everyone talks to one **JSON API**; the API reads and writes one **database** and one **media store**; and a small set of **services** handle email, SMS, push notifications, payments and maps.

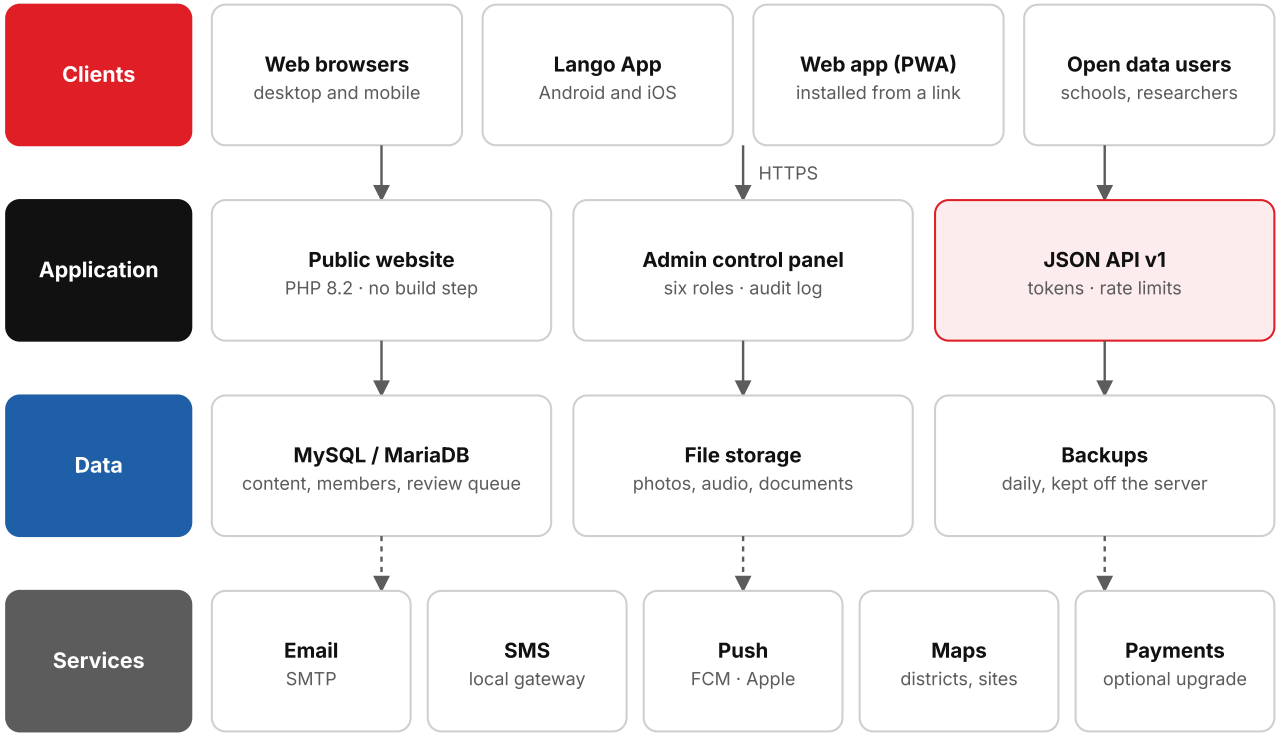


Figure 8.1: Platform architecture. Three surfaces share one API, one database and one media store. Services are connected behind the API.

8.2 Technology choices

Layer	Choice	Why
Website and admin	PHP 8.2, MySQL or MariaDB, no framework and no build step	Runs on affordable shared, VPS or cloud hosting; easy for local developers to maintain; the same approach as several of our production systems
API	Stateless JSON API (`/api/v1`) with bearer tokens, paging, rate limits and documented errors	Any future channel can use it; no sessions or cookies to manage
Mobile app	Expo SDK with React Native, Expo Router, one JavaScript codebase	One codebase for Android, iOS and the web app; fast iteration; over-the-air updates
Web app (PWA)	Built from the same app code and served from the website	No store needed; installable from a link
Push notifications	Expo Notifications over Firebase Cloud Messaging (Android) and Apple Push (iOS)	Free to start; reliable
Media	Files stored on the server with image optimisation; optional move to cloud storage	Simple now; scalable later
Maps	Interactive maps for districts, heritage sites and sightings	Visual exploration of the sub-region
Email and SMS	SMTP email; SMS through a local gateway such as Africa's Talking	Receipts, confirmations and alerts; Kakebe already integrates both in other products
Payments (optional upgrade)	ioTec Pay for MTN and Airtel mobile money and cards; Pesapal as an alternative	Both are in production in Kakebe systems (Chapter 10)

Kakebe uses this stack, or its Laravel counterpart, across its own systems, including BetterLife International, iRecover, ObiFunds, KPOS, SignConnect Uganda and FaithFellows. The technology is not experimental, and the Foundation will not be dependent on a single developer to keep it running.

8.3 Hosting and environments

Option	Description	Best for
A. Managed by Kakebe	Kakebe provisions and manages hosting on a reliable provider, with monitoring, backups and security updates under a support plan	Foundations that want to focus on content, not servers (recommended)
B. Foundation-owned hosting	The Foundation holds the hosting account; Kakebe deploys and maintains it	Foundations that want direct control of infrastructure
C. Shared hosting (Option A scope only)	A shared hosting plan suitable for the website and admin alone	Budget-conscious launch, with the option to upgrade

Whichever option is chosen, we set up a **staging environment** for testing and review and a **production environment** for the public. HTTPS is mandatory (it is also required by Android for the app). The domain, hosting account and store accounts are registered in the Foundation's name. [TBC: preferred hosting location and domain name, for example a .ug or .org address.]

8.4 Security

We combine what the platform already does with a pre-launch review.

Control	Status	Detail
Password storage	In the demonstrator	Passwords are hashed with a modern one-way algorithm; never stored in readable form
Token security	In the demonstrator	API tokens are random, and only a SHA-256 hash is stored, so a database leak does not expose live tokens
No session cookies on the API	In the demonstrator	The API is stateless, so cookie-based attacks do not apply to it
Role-based access	In the demonstrator	Six roles; clan leaders limited to their own clan; contributors have no admin access
Rate limiting	In the demonstrator	Sign-in, registration, uploads, translator and assistant requests are rate limited
Duplicate-safe writes	In the demonstrator	Uploads carry a unique reference so retries cannot create duplicates
Forced first-login password change	In the demonstrator	The first administrator must choose a new password
Audit log and version history	In the demonstrator	Who did what and when; mistaken edits can be restored
Secure token storage on the phone	In the demonstrator	Keychain on iOS, Keystore on Android
Account deletion	In the demonstrator	Members can delete their own account and remove their data link
Input handling and file uploads	Reviewed at hardening	Validation of every field, restricted file types and sizes
CSRF protection on forms	In the demonstrator	Forms carry a per-session token
HTTPS and security headers	Added at launch	Enforced on production
Dependency and configuration audit	Added at launch	Review of libraries, server settings and secrets
Penetration test	Added before launch	Our cybersecurity service tests the site and API against common attack types (the OWASP Top 10)
Backups	Added at launch	Section 8.7

8.5 Data protection and privacy

The platform will hold personal data about members and contributors: names, contact details, clan, location and recordings. We treat that as a duty of care and align the design with the **Uganda Data Protection and Privacy Act, 2019**.

Requirement	How we address it
Lawful basis and consent	Clear, plain-language consent at registration and on contribution forms, with a privacy notice in English and Leb Lango
Data minimisation	Only the fields needed are collected; phone, country and clan are optional where possible
Access, correction and deletion	Members can view and update their profile and delete their account; administrators can export or erase a person's data on request
Purpose limitation	Personal data is used for running the platform and the Foundation's work, not sold or shared with third parties
Security safeguards	Section 8.4; encrypted connections; restricted admin access
Retention	A retention schedule agreed with the Foundation, for example for unconfirmed accounts and rejected submissions
Registration and accountability	We advise the Foundation to register with the Personal Data Protection Office and to nominate a data protection focal person [TBC: confirm with the Foundation's legal adviser]
Children	Clear rules on photographs and recordings of children, and on the age of sign-up
Cross-border hosting	The hosting location is chosen and disclosed deliberately, with the lawful basis for any transfer

The demonstrator's Privacy and Terms pages are drafts for legal review, and we will work with the Foundation's legal adviser to finalise them before launch.

8.6 Performance and reliability

- **Light pages and images,** with sensible caching and compressed media, so pages load on mid-range phones on a 4G connection.
- **A lightweight reading mode** for slow connections.
- **App caching:** screens show saved data instantly and refresh in the background.
- **Monitoring** of uptime and errors under the support plan, with alerts to the Kakebe support team.
- **Capacity.** The platform comfortably serves the Foundation's expected audience on a modest server. Growth beyond that is handled by moving to a larger plan, not by rebuilding.

8.7 Backups and recovery

Item	Approach
Database	Automated daily backups, kept off the main server, plus an on-demand SQL backup from the admin panel
Uploads	Regular copies of photographs, audio and documents to separate storage
Restore test	A restore drill before launch and then periodically
Target recovery	[TBC: proposed recovery point and recovery time targets, for example no more than 24 hours of data and one working day to restore]

8.8 Integrations

Integration	Purpose	Phase
Email (SMTP)	Event tickets, contribution status, receipts, newsletters	Launch
SMS gateway	Alerts, one-time codes, event reminders	Optional, post-launch
Mobile money and cards	Donations, event tickets, sponsorships	Optional upgrade
Push notifications	Contribution decisions, events, word of the day	Launch (decisions), later (campaigns)
Maps	Districts, heritage sites, sightings	Launch
Analytics	Privacy-conscious usage insight	Launch
Open data API	Researchers, schools and partners	Launch

8.9 Scalability and the road ahead

The platform is **modular**. A new managed content type is added by registering one entry in the admin, and the API and app follow. The same backend can later feed an SMS or USSD service, a community forum, kiosks in schools and museums, partner portals, radio programmes and more, without changing the foundation.

8.10 Accessibility and inclusion

- WCAG 2.1 AA as the target for the website: sufficient contrast, keyboard navigation, labelled forms and alternative text.
- Large touch targets and readable type in the app.
- Plain language, icons with text labels, and Leb Lango alongside English.
- Offline and lite modes for people with limited data.
- Assisted contribution: a youth ambassador can submit on an elder's behalf, with the elder credited and consent recorded.

CHAPTER 9

About Kakebe Technologies

A youth-led technology company in Lira City that builds software, AI and robotics for real local problems, and trains the next generation to build them too.

9.1 Who we are

Kakebe Technologies Limited is a youth-led technology hub based in **Lira City**, Uganda, in the heart of Lango. We build software, AI tools and robotics that solve real local problems, and we train young people to build the next ones themselves. For more than five years we have worked with non-profits, government offices, schools, churches, businesses and community groups across Northern Uganda and beyond.

Item	Detail
Mission	We build homegrown software, AI and robotics, and teach young people to build the next ones themselves
Vision	A Northern Uganda where youth-led innovation solves local problems, and Lira City is known for digital talent
Office	Rm B1, Plot 43, Renzo City House, Obote Avenue, Lira City, Uganda
Phone	+256 777 676 206 · +256 393 249 845
Email and web	service@kakebe.tech · www.kakebe.tech

9.2 Our track record in numbers

Measure	Detail
5+ years	Building technology in Lira City
257 youth	Graduated from Kakebe Tech Camp 2025 alone, from across East Africa
6 service lines	Software, digital marketing, robotics and AI, cybersecurity, tech training, internships
20+ platforms	Websites, portals, ERPs, apps and SaaS products delivered or incubated (Chapter 10)
Mobile money in production	Live integrations with mobile money and card payments across several products

9.3 What we do

Service line	What it covers	How it applies to this project
Software development	Custom web and mobile apps, APIs, e-commerce	The website, admin, API and app
Digital marketing	SEO, social media, content strategy, analytics	Launch campaign, discoverability, audience growth
Robotics and AI	Automation, machine learning, IoT, smart devices	The source-bound Cultural Assistant and translator; future speech and language tools
Cybersecurity	Network security, threat assessment, audits, data protection	Security review and penetration testing before launch; data protection design
Tech training	Robotics, software and AI for youth and businesses	Training the Foundation's team, and a digital skills pipeline for Lango youth
Internship programmes	Mentorship, real projects, career preparation, certification	Supervised interns who help with content entry and digitisation (Option C)

9.4 Our programmes

Kakebe does not only deliver projects. It builds capacity in the community where the Foundation lives.

- **Kakebe Tech Camp.** A residential bootcamp for ages 13 to 30 in robotics, AI and machine learning, web and app development, entrepreneurship, engineering and finance, and multimedia. Camp 2025 trained 257 young people from across East Africa, who left with working prototypes. The next camp runs on 14 to 23 December 2026.
- **Digital Bridge Internship Programme.** Interns in IT, business and animation work alongside our engineers, designers and marketers on live projects and graduate with a certificate and a portfolio.
- **Innovation Programme.** Mentorship, technical support and incubation for young innovators, including the **Lango Innovation Challenge** and regular Tech Talks. Several of our own products began here.

For the Foundation these programmes are an asset: a pool of trained local talent, a route for interns to contribute to the platform under supervision, and a platform story that inspires the next generation.

9.5 The team

Name	Role	What they bring to the project
Sedrick Otolo	Team Lead	Leads the team and the client relationship; accountable for delivery
Patrick Okello	Chief Technology Officer	Sets technical direction, architecture and quality
Oscar Jerome Okello	Web Development Lead	Leads the website and admin build
Komakech Moses	Sales and Marketing	Regional growth and community engagement; leads stakeholder and launch communications
Bodo Desderio	SEO Engineer	Search visibility, content structure and analytics

A dedicated project team is named in Chapter 12, including the mobile developers, designer and quality assurance engineer who work with these leaders.

9.6 Why Kakebe is the right partner for the Foundation

1. **We are local.** Our office is in Lira City. We can sit with elders, clan leaders and language experts in person, run workshops in Lira and visit the districts, without flights, time zones or invoices for travel from abroad.
2. **We have already built it.** The demonstrator in Appendix B is a working version of what this proposal describes. The Foundation can see and test before committing.
3. **We have proven delivery across the same technologies.** Content-managed websites, multi-role admin systems, mobile apps on a shared API, offline-first design, mobile-money payments, event ticketing, tracking codes and government work: each appears in our portfolio (Chapter 10).
4. **We build capacity, not dependency.** The Foundation's staff are trained to run the platform, the documentation is complete, and the code, content and data belong to the Foundation.
5. **We keep the money and the skills in Lango.** The work is done in Lira by a Lira-based team and local trainees, which is consistent with a cultural institution's mission.
6. **We are accountable.** Four clear stages, fixed milestones, a staging site to review at any time, and a named team (Chapter 11).

CHAPTER 10

Portfolio and References

Platforms we have designed, built and operated, and exactly what each one proves about our ability to deliver the Foundation's website and app.

10.1 How to read this chapter

A proposal is only as credible as the work behind it. This chapter presents the reference projects most relevant to the Foundation, grouped by the kind of capability they demonstrate, followed by a matrix that maps each Foundation requirement to the live systems where we have already delivered it. Where a system serves confidential or government users, we describe it in words and offer a private demonstration with the client's permission. [TBC: Kakebe to confirm which clients agree to be named as referees.]

#	Reference	Type	What it proves for the Foundation
1	**Lango cultural platform** (demonstrator)	Website, admin, mobile app	The complete solution, already working
2	**A Better Life International**	Non-profit website, shop, admin	Foundation-style site with a full CMS and online payments
3	**Betty Amongi Foundation**	Foundation website, CMS, mentorship portal	A Lango-region foundation, content management and community programmes
4	**Pure Wage Foundation**	Foundation platform	[TBC]
5	**Civic Coded**	Civic technology platform	[TBC]
6	**OPM Schedules**	Government scheduling platform	Government-grade, multi-office, role-isolated administration
7	**iRecover**	Lost-documents recovery platform	Tracking codes, matching workflow, mobile-money payments
8	**BodaERP**	Rider registry and revenue ERP	Five-role system spanning riders to government
9	**FaithFellows**	Web platform and mobile app	Mobile app on a shared API, offline, multilingual
10	**Soila**	AI app for farmers	AI and mobile product for rural users
11	**Better Life Farmers Connect**	Farmers' application	[TBC]
12	**Lira City website**	Government website and CMS	Public institution CMS with citizen reporting
13	**SignConnect Uganda**	Sign-language learning app	Language learning with video, lessons and classes
14	**ObiFunds, ObiTickets, ChamaFunds**	Fundraising and ticketing	Mobile-money payments, QR tickets, payouts
15	**Kakebe Tech Camp platform**	Registration, payments, portal	Registration, receipts, tickets and a participant portal

#	Reference	Type	What it proves for the Foundation
16	**KPOS and Falcon ERP**	SaaS and operations ERP	Multi-tenant SaaS, multilingual, geotagged workflows

10.2 Closest reference: the Lango cultural platform

Type: Public website, admin control panel, JSON API and mobile app · **Status:** working demonstrator

The platform described in this proposal exists today as a tested demonstrator: more than 30 website page types, 30 managed content types in the admin, about 50 app screens, a 32-table database and a documented API (Chapter 1 and Appendix B). It demonstrates offline-first contribution, a moderated review queue, a Leb Lango dictionary and translator that never invent content, and bilingual interface management by language experts.

Why it matters. It removes the single biggest risk in a project of this kind: discovering late that the idea is harder than expected.

10.3 Foundations and community organisations

A Better Life International

Website, e-commerce and admin dashboard · Non-profit working on climate action, youth empowerment and sustainable development · betterlifeint.org

What we built. A full redesign of the organisation's website in a fresh green-and-blue nature theme, with real content from the live site, plus a new **BetterLife Farm** section selling honey, ghee and yoghurt. A complete admin dashboard lets staff manage programmes, farm products, team and board, testimonials, impact counters, projects, impact reports (uploaded PDF or link), a rich-text blog with categories and drafts, orders, messages and newsletter subscribers, and site settings, all without touching code. Customers add products to a cart and pay by card or mobile money through Pesapal's hosted checkout, with order confirmation, an admin alert and a paid receipt sent by email, plus printable invoices.

Relevance. The same pattern as the Foundation's site: a content-managed foundation website with a team page, programmes, impact figures, a blog, forms, newsletter, and online payments, edited by non-technical staff.

Betty Amongi Foundation

Foundation website, content management area and mentorship portal · Supports youth, women and entrepreneurs in the Lango sub-region and Northern Uganda through seed funding, mentorship and training

What we built. A professional website covering the foundation's work, entrepreneurs, training, services, testimonials, blog and gallery, with a content management area for posts, pages, gallery, comments and messages. A separate **mentorship portal** has application and login flows and distinct areas for participants, mentors and administrators.

Relevance. A foundation serving the same Lango community, with the same needs: a public story, content management, a gallery, programmes and a structured way for the community to take part.

Pure Wage Foundation

[TBC: type of work, for example website, donation platform or CMS] · *[TBC: URL]*

What we built. [TBC: two or three sentences on scope and features.]

Relevance. [TBC: for example, a foundation platform with donor and programme features.]

Civic Coded

[TBC: type of work, for example civic technology platform, community site or training programme] · *[TBC: URL]*

What we built. [TBC: two or three sentences on scope and features.]

Relevance. [TBC: for example, civic engagement and community participation.]

10.4 Government, public-service and civic systems

OPM Schedules (Office of the Prime Minister scheduling application)

Multi-tenant scheduling platform for ministry offices of the Government of Uganda

What we built. A multi-tenant platform in which each ministry office manages its weekly meetings and trips, completely isolated from every other office. Features include meetings and trips with date, time, venue, agenda, attendees and accompanying staff; automatic duration; a Monday-to-Friday weekly view with colour-coded warnings for upcoming items; clash flags that warn but never silently block; printer-friendly schedules; **branded email reminders** ahead of each meeting; and a one-click **weekly programme sent as a PDF** to every registered user in that office. A separate **Super Admin** area onboards new ministries, manages users across all offices and resets passwords. The reference office for the build was the Office of the Minister for Relief, Disaster Preparedness and Refugees.

Relevance. Government-grade role separation (office admin versus super admin), scheduled notifications, branded PDFs and multi-office data isolation, the same discipline the Foundation needs when clan leaders, editors and moderators each see only what they should.

Lira City website

Government website and CMS · A rebuild of the Lira City website

What we built. A clean rebuild of the city's public site on PHP 8 and MySQL, with an administrator panel for news, events, pages (vision, mission, mandate, history, geography), downloads, tenders and notices, FAQs, departments, technical staff, an organogram, service categories and services, a messages inbox and subscribers with CSV export. Citizens can **report an issue and track it with a tracking code**; staff run a status workflow with a public update and internal notes.

Relevance. A public institution in Lira with a large content inventory, a citizen feedback loop and tracking codes, all of which appear in the Foundation's contribution engine.

iRecover

Lost-documents recovery platform · web, with station and admin dashboards · [TBC: public URL]

What we built. A platform that helps people in Uganda recover lost official documents (National IDs, driving permits, passports, student IDs, academic documents, land titles, birth certificates and more) that have been found by partner radio stations, police posts and other collection points. A member of the public can **search** the found-documents database, **report** a lost document, get **matched**, pay a small recovery fee by **MTN or**

Airtel mobile money, and collect the document from the station that holds it. Stations upload found documents and confirm hand-over; administrators approve matches and manage fees and alerts. A public **status tracker** and a **receipt with a pickup code** close the loop.

Relevance. A community-powered, multi-party workflow with partner organisations, tracking codes, matching, mobile-money payments and role-separated dashboards, much like the Foundation's contribution, review and archive pipeline.

BodaERP

Rider registry, fee collection and compliance platform for boda-boda transport · currently in development

What we built. A platform in which **riders, stage chairpersons, city councils, government and platform administrators** each have their own workspace. Chairpersons register riders, collect payments and track defaulters and expiries; city councils monitor compliance, take enforcement actions and analyse revenue; government sees a national registry with comparisons and reports; super administrators manage cities, revenue splits, data quality, audit logs and backups. Riders see their ID card, payment history and renewal status. Mobile-money payments run through ioTec Pay, and anomaly scans flag suspicious data.

Relevance. Five-role access, digital ID cards, notifications, audit logs, data-quality controls and large-scale registries are close cousins of the Foundation's clan register and member records.

10.5 Mobile and AI products

FaithFellows (web platform and mobile app)

Multi-tenant church management and engagement platform · web, mobile app (iOS and Android) and PWA · by Kakebe Technologies

What we built. A platform in which every church gets a website, sermons (audio and video, with offline listening), a Bible with notes, live streaming with audio-only mode, events with RSVPs, paid tickets and QR check-in, giving through mobile money and cards, a moderated prayer wall, small groups, pastoral care, and notifications by in-app, email, push, SMS and WhatsApp. The **Laravel** web platform and the **Expo / React Native** mobile app share one backend and one JSON API. The platform runs in **English, Luo, Luganda and Swahili**, installs as a progressive web app, and offers a lite mode for slow connections.

Relevance. This is the architecture of the Lango App in production: one backend serving a website and a mobile app, multilingual content including a Nilotic language, offline use, push notifications and community moderation.

Soila

AI app that helps farmers test soil and grow better · in development

What it is. An AI-assisted mobile product for smallholder farmers, one of Kakebe's own products, currently in development.

Relevance. Experience with AI features and with designing for rural users on mobile.

Better Life Farmers Connect

[TBC: type of work, for example farmers' mobile application] · *[TBC: URL or store link]*

What we built. [TBC: two or three sentences on scope and features.]

Relevance. [TBC: for example, a community mobile app for farmers, linked to A Better Life International.]

10.6 Language, learning and community knowledge

SignConnect Uganda

Mobile-first web app teaching Ugandan Sign Language to Deaf children and their families

What we built. Short, segmented sign videos uploaded and tagged by an admin, with front and side views, slow motion and repeat, captions, and a learning path with stars, streaks and badges. Teachers create classes with a join code and assign lessons; parents manage child profiles and learn together. The admin has bulk upload, theme and lesson management, and search analytics including searches with no results. The design targets WCAG 2.1 AA, with large touch targets, high-contrast and larger-text options, and optional offline lessons.

Relevance. A language-learning product built around a community's own language and its teachers, with media-rich lessons, progress tracking and accessibility. It is the nearest cousin to the Foundation's Language Hub.

NursesPro Academy

E-learning platform for nursing and midwifery students in Uganda

What we built. Roles for super admin, tutors and students; a curriculum of modules and notes; classes; subscription payments with access that expires; and real hashed-password sign-in.

Relevance. Structured learning content, role-based access and payments.

10.7 Payments, ticketing and community finance

ObiFunds

Crowdfunding platform · www.obifunds.com

Campaigners raise money for personal, medical, education and community causes. Donors give by mobile money or bank. An admin team verifies campaigns and documents and releases funds. Payments in and out run through ioTec Pay, with PDF receipts, charts and transparent campaign pages.

Relevance. Donation flow, verification workflow, transparency, receipts and payouts, which are the building blocks of the Foundation's giving and sponsorship features.

ObiTickets

Event ticketing platform

Organisers list events and sell tickets; buyers pay by MTN or Airtel mobile money or card and receive QR e-tickets instantly; organisers scan tickets at the gate; an admin team reviews events and releases payouts.

Relevance. Directly reusable for Foundation cultural festivals, with a gate scanner and instant e-tickets.

ChamaFunds

Community fundraising platform. Fundraising campaigns with donation pages, Pesapal payments, downloadable receipts and withdrawals.

Kakebe Tech Camp platform

Registration, payments, receipts and participant portal

Online registration with track selection, package pricing, pay-now or pay-later with part payments, mobile money and card through ioTec Pay, **PDF receipts** by email, a camp **ticket with photo and QR code**, a participant portal with one-time email codes, sponsor-a-child payments, and an admin control panel with finance, reminders, payments ledger and exports.

Relevance. Registration with payments, tickets, a portal and finance reporting, in production for a real event.

10.8 SaaS and operations systems

KPOS (Kakebe Point of Sale)

Cloud-based, multi-tenant point of sale for Ugandan businesses, with MTN MoMo, Airtel Money, card and bank payments, inventory, purchasing, finance reports exportable to PDF, Excel and CSV, roles and audit log, REST API with keys and signed webhooks, daily backups, an installable web app, dark mode, and an interface in **English, Luganda and Kiswahili**.

Falcon ERP

Export and trade operations for Falcon Beverages (U) Ltd: a four-role pipeline (merchant, supplier, client, driver) from request to delivery, with **geotagged checkpoints**, a goods-received note with photo and hand-drawn signature, and generated and verified documents.

Relevance of both. Multi-tenant SaaS, role-based workflows, offline-capable web apps, multilingual interfaces, PDF and spreadsheet reporting, and field data captured with GPS, photos and signatures.

10.9 Selected additional work

Project	What it is
Mwonya Streams (mwonya.com)	A streaming platform for local entertainment and culture, preserving and promoting indigenous voices
Kakebe Shop Online (kakebeshop.com)	An online shop connecting Northern Ugandan businesses and customers
Oroma TV	News, shows and culture for Northern Uganda, with a content management admin, live streams and previous shows
Innovate2Day	An opportunity hub for young African innovators: verified grants, fellowships and calls, innovator profiles and an AI-assisted admin that writes and shares listings
NUFA and the NUFA Awards	Websites of the Northern Uganda Filmmakers Association and its annual awards: news, awards, partners, gallery, subscribers and an admin
Mighty Rides	Website, customer portal and staff workspace in Next.js with an accounting back office (Laravel)

Project	What it is
Kakebe Management System (KMS)	Internal platform for people, programmes, projects, learning, finance, content, reporting and impact, with monitoring and evaluation
Es Shemen International School	School website for a Christ-centred international school in Lira City
Kiryandongo High School	School website for a mixed day community school in Bweyale
H. Tubman Solutions	Website for a CCTV, security and ICT solutions company
Bellan Contractors and Suppliers	Corporate website and company profile
Nilotica Shea	Website for a premium Ugandan shea butter brand
The Big Five Heritage Collection	Website for a national movement uniting conservation, leadership, tourism, culture and national pride
Denise Ayebare	Personal advocacy website for a climate activist and youth empowerment advocate

[TBC: Kakebe to confirm the final list and any client permissions before this proposal is sent.]

10.10 Capability matrix: what the Foundation needs, and where we have already delivered it

Foundation requirement	Delivered in
Content-managed website with a no-code admin	A Better Life International, Betty Amongi Foundation, Lira City website, NUFA, Es Shemen, Innovate2Day, Lango platform
Multi-role access and approvals	OPM Schedules, BodaERP, iRecover, FaithFellows, Lira City website, Lango platform
Mobile app (Android and iOS) sharing one API with the web	FaithFellows, Lango App
Installable web app and offline use	FaithFellows, SignConnect Uganda, KPOS, Lango App
Mobile money and card payments	ObiFunds, ObiTickets, iRecover, BodaERP, KPOS, Kakebe Tech Camp, A Better Life International (Pesapal), ChamaFunds
Events and ticketing (with QR check-in)	ObiTickets, Kakebe Tech Camp, FaithFellows; events with printable tickets in the Lango platform
Public reporting with tracking codes	Lira City website, iRecover, Lango platform
Multilingual interface including local languages	FaithFellows (Luo, Luganda, Swahili), KPOS (Luganda, Kiswahili), Lango platform (Leb Lango)
Media: audio, video, galleries, live streaming	FaithFellows, SignConnect Uganda, Oroma TV, Mwonya Streams
Learning content, lessons and progress	SignConnect Uganda, NursesPro Academy, Kakebe Tech Camp, Lango platform
Dashboards, PDF and spreadsheet reports	KPOS, ObiFunds, BodaERP, OPM Schedules, KMS
Email, SMS and push notifications	FaithFellows, KPOS, OPM Schedules, iRecover, Lango App

Foundation requirement	Delivered in
Government and institutional clients	OPM Schedules, Lira City website, BodaERP
GPS, photos and signatures captured in the field	Falcon ERP, Lango App
AI features	Soila, Innovate2Day AI Writer, Lango Cultural Assistant and translator

CHAPTER 11

Delivery Approach and Project Plan

How we turn this proposal into a launched platform: four clear stages, short build cycles, a staging site the Foundation can review at any time, and fixed gates before anything goes public.

11.1 Our approach

Every Kakebe engagement, large or small, moves through the same four stages. For the Foundation they unfold as follows.

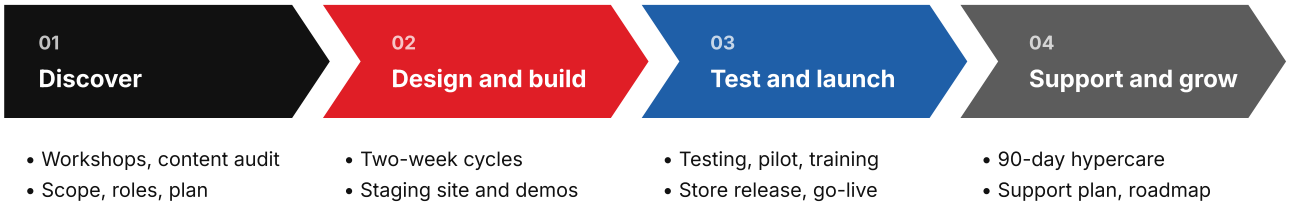


Figure 11.1: Kakebe's four-stage process applied to the Foundation's platform.

Stage	What we do	What the Foundation sees
1. Discover	Listen first. Workshops with leadership, language experts, historians, clan leaders and youth; content audit; confirm scope, roles and content rules	A signed-off scope, content inventory, role matrix and project plan
2. Design and build	Brand and interface design; build in two-week cycles with a demonstration at the end of each; content structure and import	A staging site that grows every two weeks; short review sessions
3. Test and launch	Functional, device, accessibility, performance and security testing; pilot with a small group; training; store submissions; go-live	A pilot report, trained staff, a launch plan and a live platform
4. Support and grow	Hypercare for 90 days, then the chosen support plan; quarterly review and roadmap	Monitoring, fixes, improvements and a growing archive

11.2 Working practices

- **Two-week cycles.** Each cycle ends with a demonstration on the staging site, and the Foundation's feedback goes straight into the next cycle.
- **A staging site from week 3.** The Foundation can log in and look at any time, rather than waiting for a "big reveal".
- **Weekly check-in.** A 30-minute project call, with a one-page status report: done, next, risks, decisions needed.
- **Monthly steering meeting.** Leadership reviews progress, budget, risks and priorities.
- **One backlog.** Every feature, bug and request is listed, visible and prioritised, so nothing is lost and scope changes are deliberate.
- **Definition of done.** A feature is done when it is built, tested on real devices, documented and accepted by the Foundation's coordinator.

11.3 Phases, deliverables and gates

Phase	Weeks	Main activities	Deliverables	Gate to move on
P0. Discover and align	1 to 2	Kick-off, workshops, content audit, stakeholder interviews, review of the demonstrator, scope and role confirmation	Signed scope, content inventory, role matrix, risk register, detailed plan	Foundation signs off scope and plan
P1. Brand, design and foundation	3 to 8	Brand and interface design, design system, site structure, content model, environments, hosting, admin roles, website shell	Approved designs, staging site with navigation, admin with roles, hosting ready	Design approval
P2. Culture and language modules	5 to 12	History, districts, clans, notable Langos, heritage, biodiversity, library, Language Hub (dictionary, translator, proverbs, lessons, greetings), Cultural Assistant, bilingual interface	Working modules with Foundation content loaded where available	Module demonstrations accepted
P3. Community engine and API	9 to 13	Contribution engine, review workflow, accounts, points and badges, notifications, API hardening, privacy and consent flows	Moderated contribution pipeline on the website, API v1 documented	Workflow walk-through accepted
P4. Mobile app	9 to 16	App design, build for all screens, offline dictionary and outbox, push notifications, web app, store accounts	Test builds on Android and iOS, web app on staging	App demonstration accepted
P5. Content, testing and pilot	15 to 18	Content loading, user acceptance testing, accessibility and performance checks, security review, pilot with 20 to 40 users from different districts	Test reports, security report, pilot report, defect list closed	Foundation signs the acceptance certificate
P6. Launch	19 to 20	Staff and moderator training, store submissions and release, production go-live, launch communications	Live website, live app, trained team, launch pack	Go-live approval
P7. Hypercare	90 days from go-live	Daily monitoring, rapid fixes, usage review, support hand-over	Hypercare report and support plan	End of hypercare review

The plan for **Option A** (website and admin only) has the same shape without P4 and takes about 14 weeks. These durations are indicative and are confirmed at the end of P0, once the content inventory and the Foundation's availability for reviews are known.

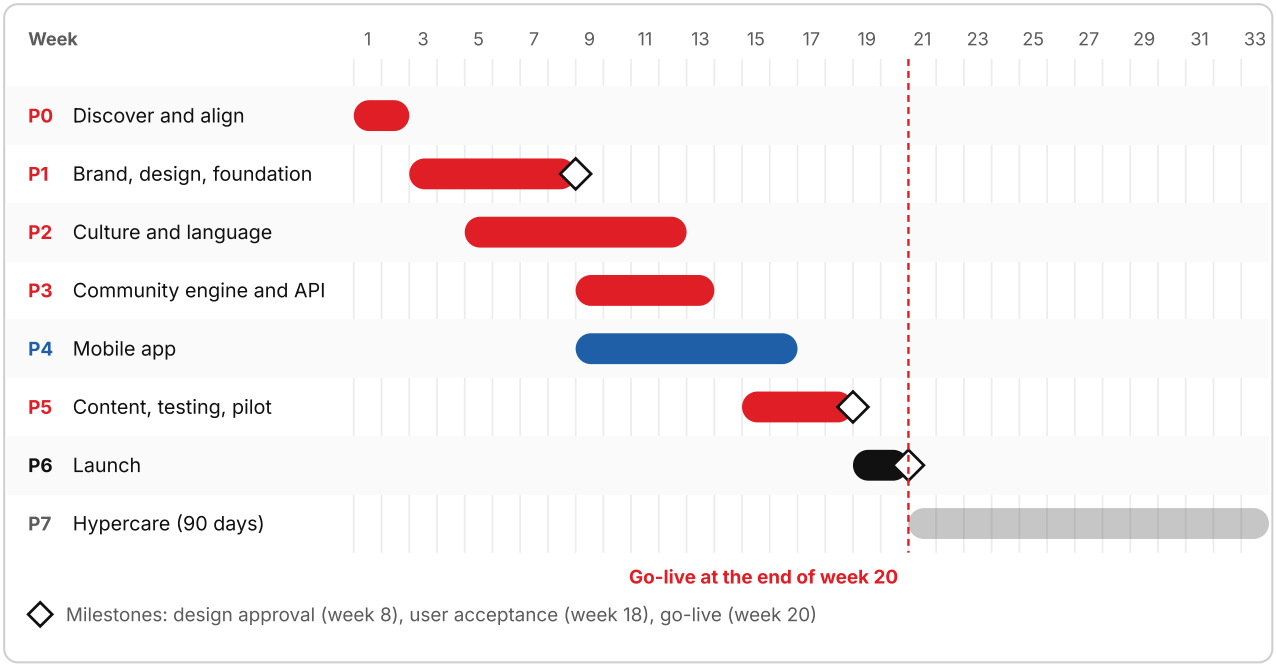


Figure 11.2: Indicative project timeline for Option B: about 20 weeks to launch, then 90 days of hypercare.

11.4 Content migration and digitisation

The platform is only as valuable as the content it holds. We plan content as a work stream, not an afterthought.

Step	What happens	Who
Audit	List existing content: documents, photographs, recordings, clan registers, dictionaries, histories; note owner, format, language and sensitivity	Kakebe with Foundation coordinator
Prioritise	Decide what must be live on launch day, what follows, and what stays restricted	Foundation
Prepare	Templates, CSV formats for clans, dictionary and proverbs; style guide for Leb Lango entries	Kakebe
Load	Bulk import (CSV) for clans, dictionary entries and proverbs; manual entry for narrative content; scanning and photography as needed	Kakebe, Foundation staff, interns under supervision
Verify	Language experts and historians review; items marked *verified* when confirmed	Foundation panels
Maintain	Ongoing contribution and review through the platform	Foundation moderators

The demonstrator ships with a starter structure and a starter dictionary marked *unverified*; the Foundation's own verified content always takes precedence and is never overwritten by updates.

11.5 Quality assurance

Test type	What we check
Functional	Every feature against its acceptance criteria
Device and browser	A range of low-, mid- and high-end Android phones, iPhones, and current browsers
Offline and low connectivity	Dictionary download, queued contributions, recovery after dropped connections
Accessibility	Contrast, keyboard navigation, labels, large text
Performance	Page weight and load time on a throttled 4G connection; app start time
Security	Dependency and configuration audit; penetration test against the OWASP Top 10
Content and language	Spelling, diacritics, bilingual consistency, broken links
User acceptance	Foundation staff and pilot users run real scenarios and record issues

Defects are classified by severity: **critical** (platform unusable or data at risk), **high** (a main function fails), **medium** (a workaround exists) and **low** (cosmetic). No critical or high defects may remain open at go-live.

11.6 Launch plan

1. **Soft launch** to the pilot group of clan leaders, language experts, youth and diaspora members, with feedback collected in a simple form.
2. **Store release.** Android first on Google Play, then iOS on the App Store (review times vary), and the web app available from day one.
3. **Public launch** with a communications pack: press release, social media assets, short how-to videos in English and Leb Lango, posters and QR codes for events, and a launch event [TBC: date and venue].
4. **Early support.** Daily monitoring and a visible support contact during the first weeks.

11.7 Change control

Requests that go beyond the agreed scope are logged, estimated in effort and cost, and approved in writing by the Foundation's coordinator before work starts. Small changes within a cycle can be swapped for equal-sized items without extra cost. Nothing is built that the Foundation has not agreed to.

CHAPTER 12

Team, Governance and Training

Who does what on both sides, how decisions are made, and how the Foundation's people are equipped to run the platform on their own.

12.1 The Kakebe delivery team

Role	Person	Responsibility
Project lead and client partner	Sedrick Otolo, Team Lead	Accountable for delivery, scope, budget and the relationship with the Foundation
Technical architect	Patrick Okello, Chief Technology Officer	Architecture, security, technical quality and code review
Website and admin lead	Oscar Jerome Okello, Web Development Lead	Website, admin, API and database
Mobile lead	[TBC: name]	App architecture, offline sync, push, store releases
Product designer	[TBC: name]	Brand application, interface design, accessibility
Search and digital engagement	Bodo Desderio, SEO Engineer	Content structure, search visibility, analytics
Community engagement and training	Komakech Moses, Sales and Marketing	Stakeholder engagement, launch communications, training coordination
Quality assurance	[TBC: name]	Test plans, device testing, defect tracking
Content and digitisation support	Interns from the Digital Bridge Internship Programme (Option C)	Supervised content entry, scanning and formatting

12.2 The Foundation team we need to work with

A small, named group on the Foundation's side is the single most important success factor. We recommend:

Role	Responsibility	Suggested time commitment
Executive sponsor	Champions the project, resolves escalations, approves milestones	A few hours per month
Project coordinator	Day-to-day decisions, gathers feedback, signs off deliverables	About 4 hours per week
Language committee	Reviews Leb Lango wording, the dictionary and translations	Scheduled review sessions
History and heritage panel (elders and historians)	Reviews history, clan and heritage content	Scheduled review sessions
Clan leader liaison	Connects clan leaders, validates the clan register	A few hours per month
Communications officer	Provides brand assets, photography and news; becomes the lead editor	About 2 days per week after launch
Finance officer	Programme and donation setup and reporting	As needed
Data protection focal person	Privacy notices, consent, requests from members	As needed

[TBC: the Foundation to nominate named individuals for each role in the first week.]

12.3 Who is responsible: a RACI summary

R = does the work · **A** = accountable · **C** = consulted · **I** = informed

Activity	Foundation sponsor	Foundation coordinator	Language and history panels	Kakebe project lead	Kakebe technical lead
Approve scope and plan	**A**	R	C	R	C
Provide brand assets and content	I	**A**	R	C	I
Design approval	**A**	R	C	R	C
Build and test	I	I	I	**A**	R
Review Leb Lango and cultural content	I	C	**A/R**	I	I
Content policy and cultural protocols	**A**	R	R	C	C
User acceptance testing	I	**A/R**	C	R	C
Security review and hardening	I	I	I	C	**A/R**
Go-live decision	**A**	R	C	R	R
Training delivery	I	R	C	**A**	R
Post-launch support	I	R	I	**A**	R

12.4 Governance and communication

Forum	Frequency	Purpose	Attendees
Steering meeting	Monthly	Progress, budget, risks, strategic decisions	Foundation sponsor and coordinator, Kakebe lead and CTO
Project call	Weekly, 30 minutes	Status, blockers, decisions	Coordinator, Kakebe lead
Cycle demonstration	Every two weeks	See the work, give feedback	Coordinator, panel members, Kakebe team
Content review sessions	As scheduled	Review language and heritage content	Panels, Kakebe
Status report	Weekly, one page	Done, next, risks, decisions	Shared by email

Escalation. Issues are raised first with the project coordinator and the Kakebe lead; unresolved issues go to the steering group within three working days.

12.5 Training and capacity building

The Foundation should be able to run the platform on its own after launch. Training is therefore part of the scope, not an extra.

Audience	Sessions	Format and length	Outcome
Administrators	Platform overview, roles and users, settings, backups, exports	1 day, hands-on, in Lira	Can manage users, settings and data
Editors and communications	News, pages, events, media, translations, SEO basics	1 day, hands-on	Can publish and maintain content
Moderators and language experts	Review queue, contribution status, dictionary and proverbs, translations	1 day, hands-on	Can review and approve contributions with confidence
Clan leaders	Clan self-service, how to correct and add information	Half day, in groups, in English and Leb Lango	Can maintain their clan page
Youth ambassadors	Assisted contribution: helping elders record stories with consent	Half day	Can record and submit on behalf of elders
Leadership	Dashboards, reports, what the platform shows	2 hours	Can use the platform for decisions and communication

Materials. An administrator manual, short video tutorials, quick-reference cards in English and Leb Lango, a moderator checklist, a style guide for entries, and a recorded training library in the admin help area.

Training of trainers. Two or three Foundation staff learn in depth, so that new staff and volunteers can be trained without Kakebe.

Post-launch clinics. Drop-in sessions, in person or online, during hypercare and at the first quarterly review.

12.6 Handover and documentation

At handover the Foundation receives:

- The **source code** and a documented deployment process, with the Foundation as owner.
- **Access** to hosting, domain, database, store accounts and push services, in the Foundation's name.
- **Technical documentation**: architecture, data model, API reference, deployment and restore instructions.
- **User documentation**: admin manual, moderator guide and training videos.
- A **content inventory** and a record of the sources behind every loaded fact.
- A **maintenance runbook** covering backups, updates and what to do when something goes wrong.

CHAPTER 13

Investment and Commercial Options

Three clear packages, a transparent breakdown, honest running costs and milestone payments tied to deliverables the Foundation can see.

Draft pricing. All amounts below are to be completed by Kakebe and confirmed with the Foundation after the discovery workshop. Currency is Ugandan shillings (UGX). The structure, not the numbers, is what this draft settles.

13.1 How we price

- **Fixed price per package,** agreed after discovery, so the Foundation knows the total before work starts.
- **Itemised.** Each major deliverable is priced separately, so the Foundation can see where the money goes and choose what to include.
- **Milestone payments** tied to approvals, not to the calendar (Section 13.5).
- **Honest running costs.** The one-time build cost and the annual cost of running the platform are shown side by side.
- **No hidden extras.** What is not included is stated in Chapter 17.

13.2 The three packages

Item	Option A: Website and admin	Option B: Complete platform *(recommended)*	Option C: Complete platform plus growth year
For whom	Foundations that want the strongest public presence first	Foundations that want to reach the whole community, in the field and abroad	Foundations that want a managed first year of growth and content
Delivery to launch	About 14 weeks	About 20 weeks	About 20 weeks, plus 12 months of managed growth
Website	Included	Included	Included
Admin control panel and roles	Included	Included	Included
Language Hub	Included	Included	Included
Community contribution engine (web)	Included	Included	Included
Mobile app and web app	Not included	Included	Included
Offline dictionary and offline contributions	Not included	Included	Included
Push notifications	Not included	Included	Included
Store publishing	Not included	Included	Included

Item	Option A: Website and admin	Option B: Complete platform *(recommended)*	Option C: Complete platform plus growth year
Training and handover	Included	Included	Included
Hypercare	90 days	90 days	90 days, then 12 months of support
Content digitisation support	Not included	Not included	Included (supervised interns)
Launch campaign and growth programme	Not included	Not included	Included
One-time investment	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]

13.3 Investment breakdown

Deliverable	Option A	Option B	Option C
Discovery, content audit and planning	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Brand application, interface design and design system	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Public website (page templates, search, bilingual)	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Admin control panel, roles and audit	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Culture modules (history, districts, clans, people, heritage, biodiversity)	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Language Hub (dictionary, translator, proverbs, lessons, assistant)	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Community contribution engine and moderation workspace	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
API and integrations (email, push, maps, analytics)	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Mobile app for Android, iOS and web	n/a	[TBC: UGX]	[TBC: UGX]
Content loading and digitisation	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Security review, penetration test and performance tuning	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Training, documentation and handover	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Launch (store publishing, go-live, launch pack)	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Growth year: support, interns, quarterly improvements, campaign	n/a	n/a	[TBC: UGX]
Contingency	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]
Total one-time investment (excluding taxes)	**[TBC: UGX]**	**[TBC: UGX]**	**[TBC: UGX]**

13.4 Annual running costs

These are the costs of keeping the platform online. They are separate from the build and are paid whichever package is chosen. We show them openly so that the Foundation can plan its budget.

Item	What it covers	Indicative annual cost
Hosting and backups	Server or cloud hosting, storage, daily backups, monitoring	[TBC: UGX per year]
Domain and certificate	Domain name registration and HTTPS certificate	[TBC: UGX per year]
Email and SMS	Transactional email; SMS gateway if used	[TBC: UGX per year, depending on volume]
Push notifications and build services	Free tiers at launch; paid tiers only if volume requires	[TBC: UGX per year]
Store accounts	Google Play developer registration (one-time) and Apple Developer Programme (annual)	[TBC: confirm current fees at the time of signing]
Payment gateway fees	Percentage per transaction, only if live payments are enabled	Per transaction
Support plan	Chapter 14	[TBC: UGX per month, by tier]

13.5 Proposed payment schedule

Milestone	Share of total	Trigger
Kick-off	[TBC: for example 30%]	Signed agreement and start of Phase 0
Design approval	[TBC: for example 25%]	Foundation approves the designs and structure (end of Phase 1)
User acceptance	[TBC: for example 25%]	Foundation accepts the platform after testing and pilot (end of Phase 5)
Go-live	[TBC: for example 20%]	Public launch and handover (end of Phase 6)

Option C's growth year is invoiced [TBC: quarterly or monthly] in advance. Payment terms are [TBC: for example 14 days] from invoice.

13.6 Terms of the quotation

- Prices are in UGX and **exclude taxes** (VAT, withholding tax) where they apply.
- The quotation is valid for [TBC: for example 30 days] from the date of this document.
- Third-party fees (store accounts, gateway fees, SMS credits, domain) are paid by the Foundation at cost and are not marked up.
- Work outside the agreed scope is quoted separately under change control (Section 11.7).
- Intellectual property terms are in Section 17.4.

13.7 Ways to fund and sustain the platform

A cultural institution should not carry the whole cost alone. The platform itself can help, and we are happy to support the Foundation in preparing for any of the following.

Source	How the platform helps
Grants for digital cultural preservation	A funded, accountable digital archive with a transparency dashboard is an attractive proposition for partners and development agencies
Diaspora giving and sponsorship	The donation, pledge and sponsor-a-programme flow reaches Langos abroad directly
Event ticketing	Festival and cultural event tickets, printable now and with QR check-in as an option
Partnerships	Visible programme pages and partner listings for businesses and institutions
Memberships (future)	Clan or supporter memberships, once live payments are enabled

We can include a short **sustainability plan** in the discovery deliverables so that the Foundation has a clear view of running costs and income options before it commits.

CHAPTER 14

Support, Maintenance and Service Levels

A launched platform needs care. Here is what is included after go-live, the support plans on offer, and the response times the Foundation can expect.

14.1 Hypercare: the first 90 days are included

For the first 90 days after go-live we watch the platform closely and fix problems fast.

- **Daily monitoring** of availability, errors and performance.
- **Rapid fixes** for defects found by real use, at no extra charge.
- **Usage review** at 30, 60 and 90 days: what people use, where they struggle, which content is missing.
- **Training clinics** for staff and moderators.
- **Hypercare report** and a handover to the chosen support plan.

14.2 Support plans

After hypercare the Foundation chooses a plan. Prices are to be completed in Chapter 13.

Item	Essential	Standard *(recommended)*	Premium
Purpose	Keep the platform safe and online	Keep it safe, online and improving	A managed digital partner
Hosting monitoring and uptime alerts	Included	Included	Included
Security updates and patching	Included	Included	Included
Daily backups and quarterly restore test	Included	Included	Included
Defect fixes	Included	Included	Included
Support channels	Email	Email, phone and WhatsApp	Email, phone, WhatsApp and a named contact
Support hours	Working days	Working days, extended hours	Working days, extended hours, urgent out-of-hours cover
Improvement hours per month	None	[TBC: hours]	[TBC: hours]
App store updates and OS compatibility	On request	Included	Included
Quarterly review and roadmap	Annual	Quarterly	Quarterly, plus monthly report
Training refreshers	On request	Twice a year	Quarterly

Item	Essential	Standard *(recommended)*	Premium
Monthly price	[TBC: UGX]	[TBC: UGX]	[TBC: UGX]

14.3 Severity levels and response targets

The following targets are proposed and are confirmed in the support agreement.

Severity	Definition	First response	Target resolution or workaround
Critical	The platform is down, data is at risk or security is compromised	[TBC: for example 1 hour]	[TBC: for example 4 hours]
High	A main function fails for many users	[TBC: for example 4 hours]	[TBC: for example 2 working days]
Medium	A function fails for some users; a workaround exists	[TBC: for example 1 working day]	[TBC: for example 5 working days]
Low	Cosmetic issue or question	[TBC: for example 2 working days]	Next scheduled release

Availability target. [TBC: for example 99.5 percent monthly, excluding planned maintenance announced in advance.]

14.4 What maintenance covers

Area	Activities
Security	Operating system and software updates, dependency updates, certificate renewal, security monitoring
Reliability	Uptime monitoring, error tracking, backups, restore tests, capacity checks
Performance	Query and image optimisation, caching reviews, load checks before big events
Mobile	Compatibility with new Android and iOS versions, store policy changes, over-the-air updates for small fixes
Content platform	Fixes to import tools, translations screen and moderation workflow
Data	Retention schedule, export on request, privacy requests

14.5 Releases and change

- **Regular releases** on a published schedule, with release notes in plain language.
- **Emergency fixes** released immediately when needed.
- **A versioned API** so that the app keeps working when the website is updated.
- **A shared roadmap** reviewed each quarter: new features, improvements and the roadmap items in Section 6.9.

14.6 Continuity and exit

We want the Foundation to stay with Kakebe because the service is good, not because it is trapped.

- The Foundation owns the code, content, data, domain, hosting accounts and store accounts.
- Documentation and a deployment runbook are handed over, so another competent developer can take over.
- Content and data can be exported at any time in standard formats (CSV, JSON, SQL).
- On termination of a support plan, Kakebe provides an orderly handover of credentials and knowledge within [TBC: for example 30 days].

CHAPTER 15

Risks and Mitigation

What could go wrong, how likely it is, how much it would matter, and what we will do about it before it does.

15.1 Risk register

We keep a living risk register throughout the project and review it at every weekly call and monthly steering meeting. These are the principal risks at the start.

#	Risk	Likelihood	Impact	Mitigation	Owner
1	**Disputed or inaccurate cultural content** damages trust	Medium	High	Source citations; trust labels; Cultural Review Panel; nothing published without approval; officeholders shown exactly as the Foundation confirms	Foundation sponsor
2	**Content arrives late or incomplete,** delaying launch	High	Medium	Content audit in week 1; launch with a defined minimum content set; bulk import tools; supervised interns; content treated as a work stream with owners	Foundation coordinator
3	**Low connectivity** limits use	High	Medium	Offline-first app, versioned offline dictionary, lite web mode, queued uploads	Kakebe technical lead
4	**Limited stakeholder availability** for reviews	Medium	High	Named roles and time commitments in week 1; scheduled review sessions; two-week demonstrations; decision log	Both coordinators
5	**Low adoption** after launch	Medium	High	Launch plan, ambassadors, clan leader onboarding, events, recognition, regular content (word of the day, news); success measures reviewed quarterly	Foundation communications, Kakebe
6	**Spam, abuse or inappropriate uploads**	Medium	Medium	Moderation before publication; rate limits; account controls; report routes	Moderators
7	**Privacy breach or data loss**	Low	High	Security controls and review; encrypted connections; backups with restore tests; data protection design and training	Kakebe technical lead
8	**Scope creep**	Medium	Medium	Agreed scope; single backlog; change control; roadmap for later phases	Kakebe project lead
9	**Key-person dependency** on either side	Medium	Medium	Named deputies; documentation; training of trainers	Both
10	**App store delays or rejection**	Medium	Medium	Early account setup; store guideline review; web app available from day one; Android first	Kakebe mobile lead
11	**Payment integration delays** (if enabled)	Medium	Low	Treated as an optional upgrade; pledges recorded manually in the first release	Kakebe project lead
12	**Hosting or infrastructure failure**	Low	High	Reputable hosting, monitoring, daily off-site backups, restore plan	Kakebe technical lead
13	**Sensitive or restricted knowledge shared inappropriately**	Low	High	Restricted-record setting, consent capture, protocol agreed with the Foundation, review tiers	Cultural Review Panel
14	**Sustainability: running costs unfunded**	Medium	Medium	Running costs shown openly (Section 13.4); funding options (Section 13.7); support plans sized to budget	Foundation sponsor

#	Risk	Likelihood	Impact	Mitigation	Owner
15	**Leadership or institutional changes** during the project	Medium	Medium	Governance by role, not individual; written decisions; neutral presentation of officeholders	Foundation sponsor

15.2 How we manage risk

- **Early visibility.** A staging site and a two-week demonstration rhythm mean problems surface in days, not months.
- **Decisions in writing.** A decision log records what was decided, by whom and when.
- **Escalation.** Risks are escalated within three working days if they are not resolved at project level.
- **Contingency.** A contingency line in the budget (Chapter 13) covers the unexpected without a dispute.

CHAPTER 16

Measuring Success

How the Foundation will know the platform is working: a short set of indicators, linked to the objectives in Chapter 2 and reviewed every quarter.

16.1 Principles

- **Few, meaningful measures.** We prefer ten that the Foundation actually uses to twenty that nobody reads.
- **Built into the platform.** The admin dashboard shows most of these numbers without extra tools.
- **Targets set with the Foundation.** The baselines and targets below are agreed in discovery; we do not impose numbers we cannot defend.
- **Privacy first.** Measures count activity, not individuals.

16.2 Indicators

Objective	Indicator	Source	Baseline	6-month target	12-month target
Preserve	Records in the archive (clans, dictionary entries, proverbs, heritage items, documents)	Admin dashboard	[TBC]	[TBC]	[TBC]
Preserve	Share of records marked *verified*	Admin dashboard	[TBC]	[TBC]	[TBC]
Preserve	Oral histories and audio recordings captured	Contribution reports	[TBC]	[TBC]	[TBC]
Teach	Dictionary searches and lesson completions per month	Analytics, API	[TBC]	[TBC]	[TBC]
Teach	Offline dictionary downloads	App analytics	[TBC]	[TBC]	[TBC]
Connect	Registered members, by district, clan and country	Admin dashboard	[TBC]	[TBC]	[TBC]
Connect	Monthly active users (website and app)	Analytics	[TBC]	[TBC]	[TBC]
Connect	App installs and web app installs	Store consoles	[TBC]	[TBC]	[TBC]
Represent	Visits from search engines; citations by media and schools	Analytics	[TBC]	[TBC]	[TBC]
Participate	Contributions per month and share approved	Contribution reports	[TBC]	[TBC]	[TBC]
Participate	Median time from submission to decision	Moderation reports	[TBC]	[TBC]	[TBC]
Sustain	Registrations, tickets and donations	Admin dashboard	[TBC]	[TBC]	[TBC]
Include	Share of use on mobile and in Leb Lango; offline sessions	Analytics	[TBC]	[TBC]	[TBC]
Govern	Staff who can run the platform unaided	Training records	[TBC]	[TBC]	[TBC]
Reliability	Uptime, number of critical incidents	Monitoring	n/a	[TBC]	[TBC]

16.3 Review rhythm

- **Monthly:** the Foundation's coordinator reviews the dashboard.
- **Quarterly:** a review meeting with Kakebe covers the indicators, what content is missing, what users ask the assistant that it cannot answer, and the roadmap for the next quarter.

- **Annually:** a report to leadership and partners, written in plain language, that can also be shared with donors.

16.4 What good looks like in twelve months

By the end of the first year we want the Foundation to be able to say, with evidence:

1. The **Lango language, clans, districts and heritage** are documented in a trusted, sourced and growing public archive.
2. **Young Langos** use the app to learn and to contribute, and **elders' knowledge** is being recorded with consent.
3. The **diaspora** is connected, informed and giving.
4. The Foundation's **own team runs the platform**, supported by a clear, affordable plan.
5. Partners, donors, schools and media treat the platform as a **reference source**.

CHAPTER 17

Assumptions, Dependencies and Exclusions

The conditions on which the timeline and price in this proposal depend, so that there are no surprises on either side.

17.1 Assumptions

#	Assumption
1	The Foundation nominates a project coordinator and the roles in Section 12.2 within the first week
2	The Foundation provides feedback and approvals within **five working days** of each request; delays extend the timeline by the same time
3	The Foundation supplies its logo, brand colours and photographs, or agrees a licensed alternative, by the end of Phase 0
4	Cultural, historical and language content is supplied or verified by the Foundation's own experts; Kakebe structures, loads and formats it
5	The platform is delivered in **English and Leb Lango**; further languages are quoted separately
6	Hosting is on a reputable provider with HTTPS; the Foundation holds the domain, hosting and store accounts
7	Live payment collection (mobile money and cards) is an **optional upgrade**, not part of the first release, unless the Foundation selects it
8	The scope matches the demonstrator, adapted to the Foundation's identity and requirements; materially different requirements are handled through change control
9	The Foundation's legal adviser reviews the Privacy and Terms pages, which we draft
10	The pilot involves 20 to 40 users and is arranged by the Foundation with Kakebe's help

17.2 Dependencies on the Foundation

- Named people for each role, available for scheduled reviews.
- Access to existing content, registers and documents, with permission to digitise them.
- Decisions on cultural protocols, restricted content, public recognition features and the Cultural Review Panel.
- Store accounts (Google Play Console, Apple Developer Programme) and a domain name, in the Foundation's name.
- Legal review of privacy notice, terms and contributor licence.
- A launch communication plan on the Foundation's channels.

17.3 Exclusions

The following are not in the price unless added by written agreement:

- Writing, researching or translating original cultural or historical content. (We can recommend people and processes, but the Foundation's authorities own this work.)

- Professional photography, audio or video production, and large-scale scanning beyond Option C's scope.
- Live payment integration and payment gateway fees.
- SMS and USSD services, an open community forum, and the other roadmap items in Section 6.9.
- Printed materials, merchandise and event production.
- Legal advice, including data protection registration.
- Hardware, devices and internet connectivity for the Foundation's offices.
- Third-party fees such as hosting, domains, store accounts and SMS credits, which are paid by the Foundation at cost.

17.4 Intellectual property and confidentiality

- **Content and data** belong to the Foundation. Contributors keep authorship of what they submit and grant the Foundation a licence as stated on the contribution form.
- **Custom code** written for the Foundation is licensed to the Foundation without time limit for its own use and modification [TBC: final licence wording to be agreed with the Foundation's legal adviser].
- **Kakebe's pre-existing components** (including the foundation code of the platform and general-purpose libraries) remain Kakebe's property and are licensed to the Foundation for use with the platform.
- **Third-party and open-source components** remain under their own licences, which are listed in the handover documentation.
- **Confidentiality.** Each party keeps the other's confidential information private; personal data is handled under Section 8.5.

17.5 Acceptance

A deliverable is accepted when it meets its documented acceptance criteria, has passed testing with no critical or high defects open, and is signed off by the Foundation's coordinator. If the Foundation does not respond within [TBC: for example 10 working days] of a delivery notice, Kakebe will escalate to the executive sponsor before treating the deliverable as accepted.

CHAPTER 18

Next Steps and Acceptance

From reading this proposal to a signed agreement and a start date, in five simple steps.

18.1 The path to kick-off

Step	What happens	Who	Target
1. Review	The Foundation reads this proposal and notes questions	Foundation leadership	[TBC: date]
2. Demonstration	A live walk-through of the demonstrator: website, admin panel and app on a phone	Kakebe	At the workshop
3. Discovery workshop	A 90-minute session with leadership and a working group of language experts, historians and communications staff to confirm objectives, scope, roles, content and timing	Foundation and Kakebe	[TBC: date]
4. Final quotation and agreement	Kakebe issues the final scope and price; both parties sign	Both	Within [TBC: 5 to 10] working days of the workshop
5. Kick-off	Phase 0 begins; coordinator and panels named; first deliverables scheduled	Both	[TBC: start date]

18.2 Questions for the discovery workshop

To use the 90 minutes well, we suggest the Foundation consider these in advance.

1. Which **package** (A, B or C) best fits the Foundation's priorities and budget?
2. Who will be the **project coordinator**, and who will sit on the **language committee** and **history and heritage panel**?
3. What **existing content** is there (registers, dictionaries, histories, photographs, recordings), and who owns and can approve it?
4. What are the **cultural protocols** for restricted knowledge, recordings of elders and photographs of children?
5. What **identity** will the platform carry: name, logo, colours, domain name?
6. How should **leadership and officeholders** be presented, and who approves changes to those pages?
7. Which **recognition features** (points, levels, leaderboard) should be public?
8. Is **live payment** (mobile money and cards) wanted at launch or later?
9. What are the **key dates** (festivals, anniversaries, funding deadlines) the launch should align with?
10. Which **partners, donors or institutions** should be involved or informed?

18.3 Acceptance of this proposal

By signing below, the parties confirm that they wish to proceed on the basis of this proposal, subject to the final quotation and agreement described in Section 18.1.

Item	For the Foundation	For Kakebe Technologies Limited
Name	[TBC]	[TBC]
Title	[TBC]	[TBC]
Signature		
Date		

18.4 Contact

Item	Detail
Kakebe Technologies Limited	Rm B1, Plot 43, Renzo City House, Obote Avenue, Lira City, Uganda
Phone	+256 777 676 206 · +256 393 249 845
Email	service@kakebe.tech
Web	www.kakebe.tech
Proposal contact	[TBC: name, title, direct phone and email]

Thank you for the opportunity to propose. We would be proud to build this with the Foundation, in Lango, for Lango.

APPENDIX A

Complete Feature Catalogue

Every feature of the proposed platform, with where it appears and how it relates to the working demonstrator.

Status key. ● **Demonstrated:** already working in the demonstrator, adapted to the Foundation's identity and content. ◐ **New:** built in this engagement. ○ **Optional or roadmap:** not in the first release unless selected.

A.1 Culture and heritage

ID	Feature	Website	Admin	App	Status
A1	Home page with quick actions, word of the day, featured people, news, events and giving call	✓		✓	●
A2	About and leadership: the Won Nyaci, cabinet and elders	✓	✓	✓	●
A3	History: chapters, table of contents, timeline, glossary and sources	✓	✓	✓	●
A4	Districts and Lira City: atlas, comparison table, map and a page per district	✓	✓	✓	●
A5	Clan Explorer and a page per clan, showing totem, district and clan leader	✓	✓	✓	●
A6	Clan self-service for clan leaders; bulk add and CSV import of clans		✓		●
A7	Notable Langos with biographies and categories	✓	✓	✓	●
A8	Heritage items: ceremonies, attire and crafts, music and dance, symbols, sites on a map	✓	✓	✓	●
A9	Biodiversity register and community sightings map	✓	✓	✓	●
A10	Digital library: published documents with downloads	✓	✓	✓	●
A11	Media centre: photo, video, audio and podcast	✓	✓		●
A12	360 degree tours	✓	✓		●

A.2 Language

ID	Feature	Website	Admin	App	Status
B1	Dictionary: both directions, plain-keyboard search, topics, A to Z	✓	✓	✓	●
B2	Offline dictionary download, versioned			✓	●
B3	English to Leb Lango translator, dictionary-based	✓		✓	●
B4	Proverbs, greetings, phrasebook and lessons	✓	✓	✓	●
B5	Suggest a word or proverb, duplicate check, review and approval	✓	✓	✓	●
B6	Trust labels: Verified, Community, Unverified	✓	✓	✓	●
B7	Interface translation by experts in the admin; the app registers its phrases	✓	✓	✓	●
B8	Audio pronunciation recordings for dictionary entries	✓	✓	✓	◐

A.3 Community and governance

ID	Feature	Website	Admin	App	Status
C1	Contribution engine: seven types, optional account creation, clan picker, GPS, uploads, tracking code	✓	✓		●

ID	Feature	Website	Admin	App	Status
C2	Guided contribution forms with camera, microphone, gallery, files and GPS			✓	●
C3	Offline outbox with duplicate-safe retries			✓	●
C4	Moderation queue: review notes, approve, return, reject, publish, add to archive		✓		●
C5	Notifications: admin bell; in-app and push to members		✓	✓	●
C6	Points, Bronze to Platinum levels, badges and leaderboard	✓	✓	✓	●
C7	Accounts: sign-up with clan, email or phone sign-in, guest browsing, delete account	✓	✓	✓	●
C8	Professionals directory with verification	✓	✓	✓	●
C9	Cultural Assistant answering only from reviewed content, with flagged-answer review	✓	✓	✓	●
C10	Cultural Review Panel tier and restricted-record setting		✓		◐
C11	Consent capture and attribution options on contribution forms	✓	✓	✓	◐
C12	Assisted contribution mode for youth ambassadors recording on behalf of elders	✓	✓	✓	◐

A.4 Engagement, events and giving

ID	Feature	Website	Admin	App	Status
D1	News and blog: categories, bilingual posts, moderated comments, share buttons	✓	✓	✓	●
D2	Events with online registration and printable ticket	✓	✓	✓	●
D3	QR ticket check-in at the gate		✓		○
D4	Programmes with progress and sponsor button	✓	✓	✓	●
D5	Donation pledges and transparency dashboard	✓	✓	✓	●
D6	Live mobile money and card payments	✓	✓	✓	○
D7	Scholarship, grant, volunteer, partner, clan and diaspora forms	✓	✓		●
D8	Newsletter subscription with exportable list	✓	✓		●
D9	Open data API and licence page	✓			●

A.5 Platform, security and delivery

ID	Feature	Website	Admin	App	Status
E1	Six roles with scoped permissions		✓		●
E2	Version history, restore and audit log		✓		●
E3	Database backup and CSV or JSON export		✓		●
E4	Site search and faceted Data Center	✓		✓	●
E5	JSON API v1: tokens, rate limits, paging, documented errors				●
E6	Installable web app (PWA)			✓	●
E7	Push notifications through Firebase and Apple Push			✓	●
E8	The Foundation's branding: name, logo, colours, imagery, app icon and store listings	✓	✓	✓	◐
E9	Privacy-conscious analytics	✓	✓	✓	◐
E10	Structured data and SEO enhancements	✓			◐
E11	Security review and penetration test	✓	✓	✓	◐
E12	Hosting, staging and production environments, monitoring, automated backups	✓	✓	✓	◐

ID	Feature	Website	Admin	App	Status
E13	Google Play and Apple App Store publishing			✓	◐
E14	Training, documentation and handover		✓		◐

A.6 Roadmap beyond the first release

ID	Feature	Value
R1	SMS and USSD fallback	Reach on basic phones and without data
R2	One-time-code (OTP) sign-in	Easier sign-in for first-time users
R3	Community forum and clan discussion groups	A moderated space for clans and the diaspora
R4	Maps and 360 degree tours inside the app	Brings places to life on the phone
R5	Extended AI assistant, still limited to reviewed content	A richer way to ask questions
R6	Kiosks and school and museum displays	Brings the archive to places without phones

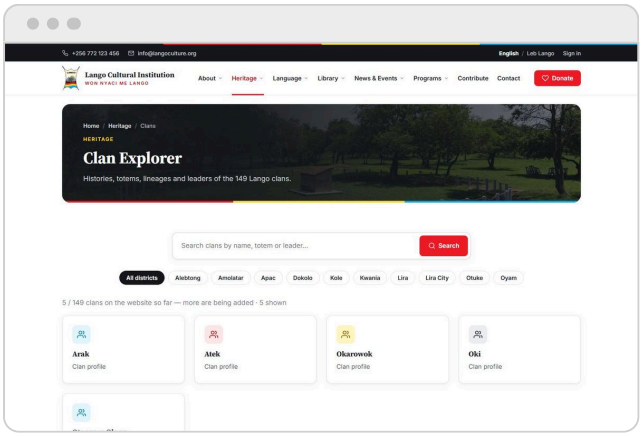
APPENDIX B

The Working Demonstrator

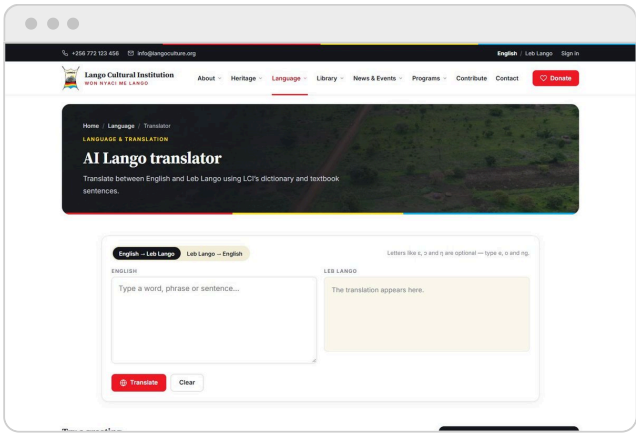
Screens from the platform as it runs today. The Foundation's own branding, content and structure will replace the demonstrator's.

All screens below are captured from the working demonstrator built under the working title *Lango Cultural Institution*. The records that appear in the admin screens are sample items created for testing; the content shown is sourced as described in Appendix C.

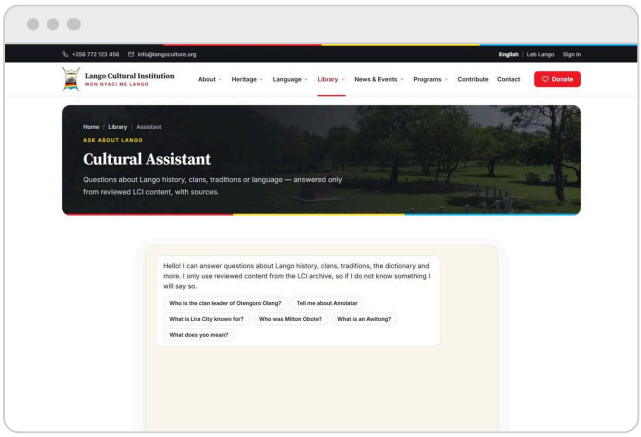
B.1 The public website



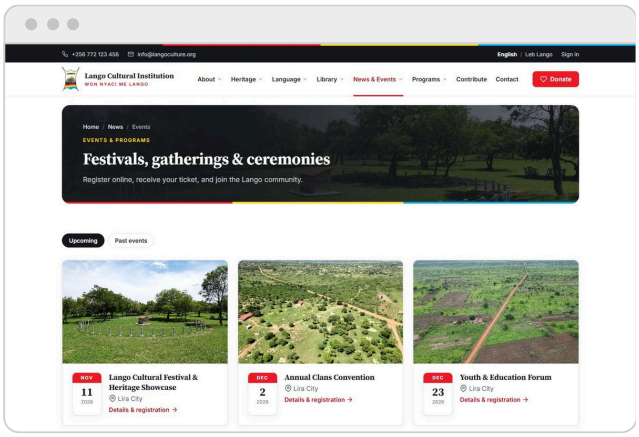
The Clan Explorer: search, filters and a page per clan.



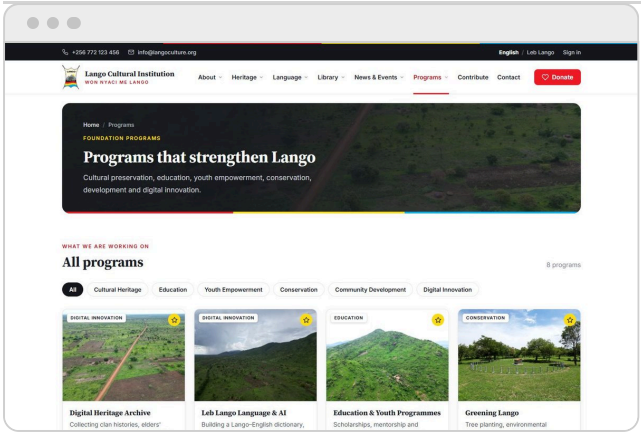
The English to Leb Lango translator, with the dictionary behind it.



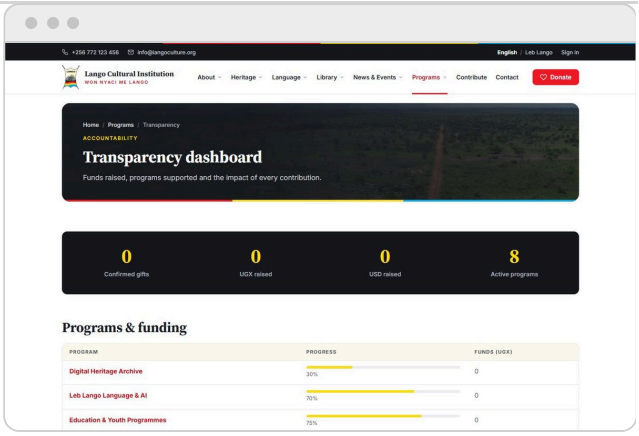
The Cultural Assistant: answers only from reviewed content, with sources.



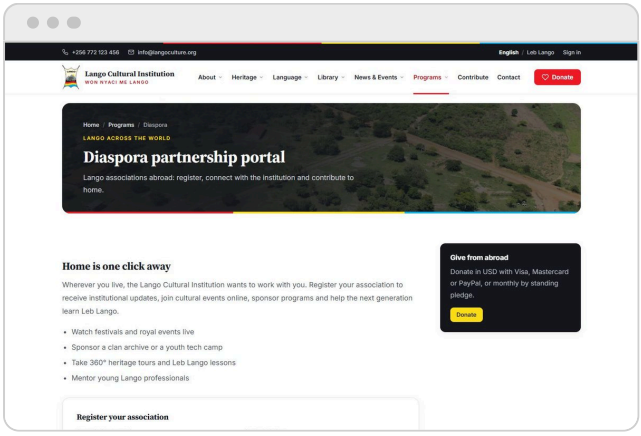
Events with online registration and a printable ticket.



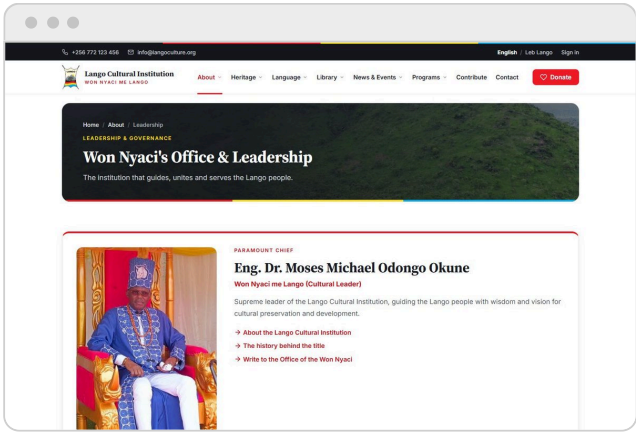
Programmes with progress and a sponsor button.



The transparency page: confirmed donations and programme figures.

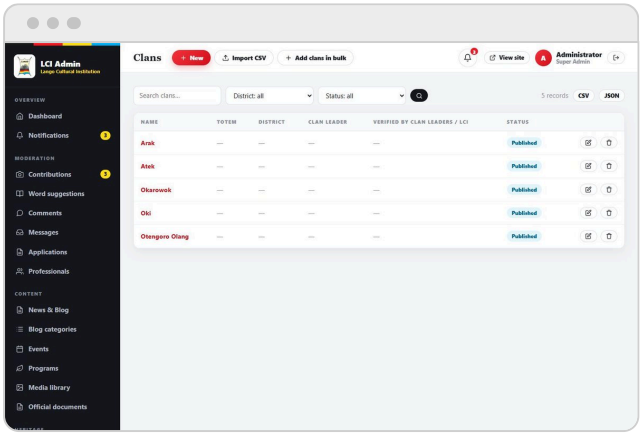


The diaspora corner.

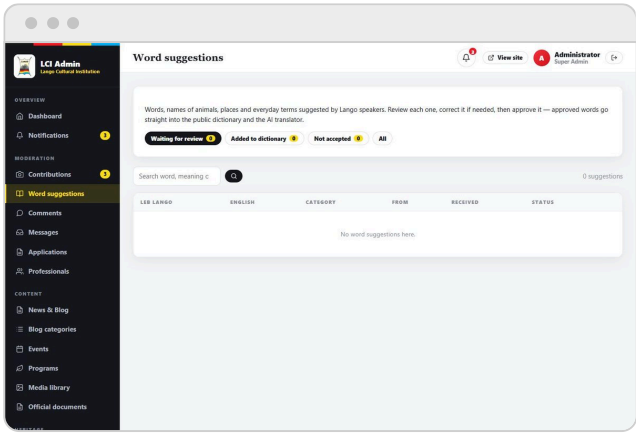


Leadership page: the Won Nyaci, cabinet and elders.

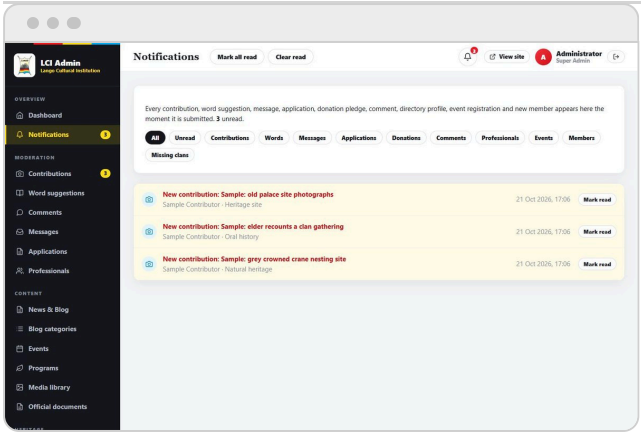
B.2 The admin control panel



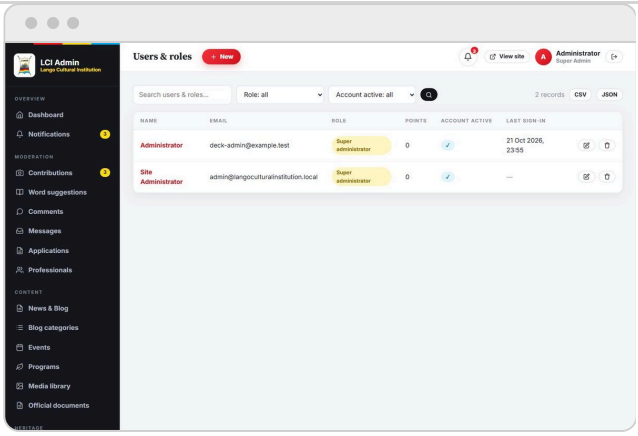
Clan management, with bulk add, CSV import and filters.



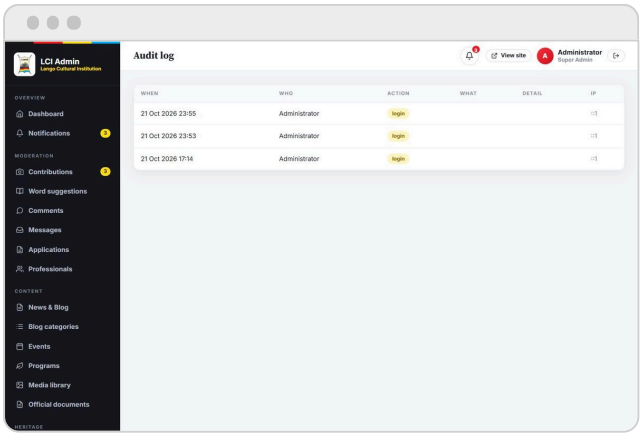
Word suggestions awaiting review, with quick actions.



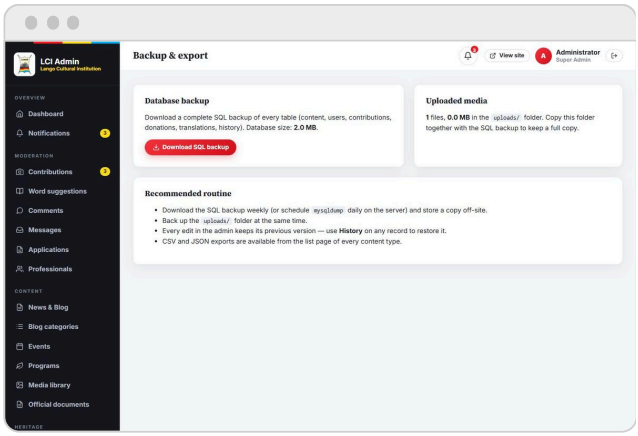
The Notifications page: every new submission in one list.



Users and roles.

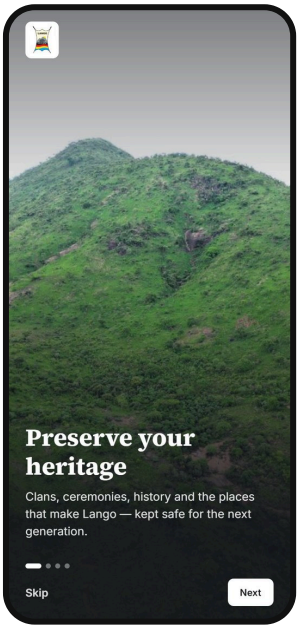


The audit log: who did what and when.

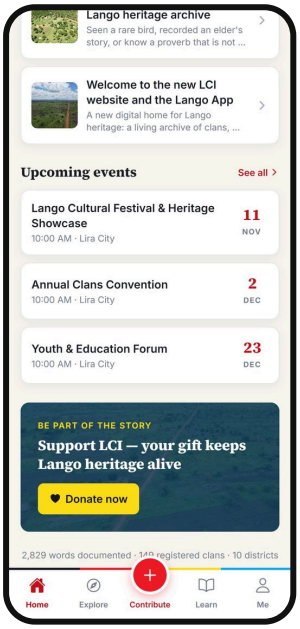


Backup and export.

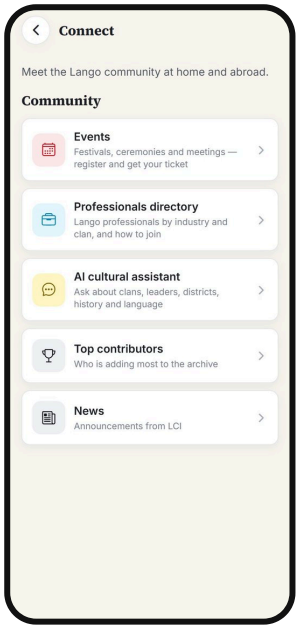
B.3 The mobile app



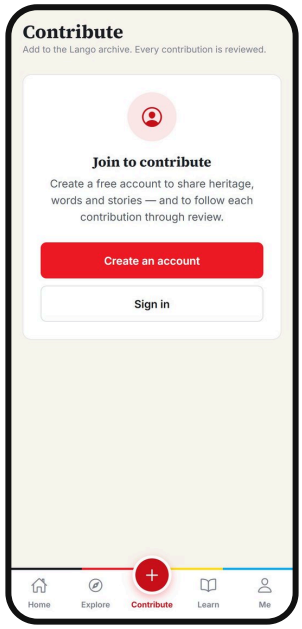
Welcome and onboarding.



Home.



Connect: events, professionals and the assistant.



Contribute: guided forms.

APPENDIX C

Content Integrity Policy

The rules that decide what the platform may publish, so that a visitor can trust every page.

C.1 The rule

Nothing cultural is invented. Every piece of content on the platform comes from one of four sources, and each is labelled.

Source	How it is handled	Label
The Foundation	Supplied or approved by the Foundation's authorities and experts	Verified
Published sources	Taken from named published works (for example official statistics, academic and press articles, encyclopaedic sources); the source is shown on the page	Source cited
Community contributions	Submitted by members, reviewed by a moderator, approved or published	Community
Starter data	Loaded to give the platform a useful start (for example the starter dictionary); flagged until experts review it	Unverified

C.2 What the platform will not do

- It will not generate clan names, proverbs, translations, biographies or histories that no person or source has supplied.
- It will not machine-translate the interface or content and present it as expert translation.
- It will not publish a contested claim, or take a side on a dispute.
- It will not publish photographs, recordings or documents without confirmed permission.
- It will not display personal data of members beyond what they have chosen to share.

C.3 What the demonstrator shows today

- **Clans.** Only clans that could be sourced are pre-loaded; the register is designed to receive the Foundation's full list by bulk add or CSV import, and new clans enter as drafts for verification.
- **Proverbs.** None are pre-loaded; the public suggestion form feeds the review queue.
- **History, districts and notable people.** Compiled from published sources, with the sources listed on the page, and flagged for the Foundation's historians to review before launch. District populations are census figures from the Uganda Bureau of Statistics.
- **Dictionary.** A starter dictionary of 2,829 entries, flagged *unverified* until experts review it. The provenance and licence of the starter data are confirmed with the Foundation during discovery.
- **Leadership.** Cabinet and elder entries are placeholders until the Foundation supplies and approves them.
- **Sample items.** News, events and programme dates in the demonstrator are placeholders to be replaced before launch.

C.4 Review before launch

Before the public launch, the Foundation's panels review: every history chapter and glossary entry; every district and notable-person profile; the dictionary entries chosen for launch; the leadership page; the privacy notice, terms and contributor licence; and every photograph and document, for permission.

APPENDIX D

API Overview

The interface that the website, the app and any future channel use. The complete reference is delivered with the project.

Group	Examples	Notes
Accounts	Register, sign in, sign out, profile, change password, delete account, my contributions, notifications, register a phone for push	Bearer token; only a hash of the token is stored
Contributions	Submit a contribution with media; check status by tracking code	Duplicate-safe through a unique reference; rate limited
Content (read only)	Home, clans, districts, people, history, heritage, species, library, news, events, programmes, transparency	Paged; English or Leb Lango on request
Language	Dictionary search; full dictionary export for offline use; proverbs; greetings; translator	The export is versioned and returns *unchanged* when nothing changed
Assistant	Ask a question	Answers only from reviewed content, with sources
Engagement	Event registration and ticket; donation pledge; professionals directory	Pledges are confirmed by the finance office
Interface text	Register phrases; fetch Leb Lango translations	Powers the Translations screen
Health	Ping	For monitoring

Conventions. JSON in and out; success responses carry the data and, for lists, paging details; errors carry a message and field errors with the usual status codes; no sessions or cookies; rate limits answer with a standard status code; cross-origin access is allowed for the web app.

APPENDIX E

Glossary

Term	Meaning
Admin control panel	The password-protected workspace where the Foundation's staff manage content, review contributions and run the platform
API (application programming interface)	The set of requests the website and the app use to read and write data
App (mobile app)	The Lango App for Android and iOS
Audit log	A record of who did what, and when, in the admin
Bearer token	A secret code that proves who is signed in to the app, stored on the phone
CMS (content management system)	Software that lets non-technical people edit a website
Contribution	A piece of knowledge submitted by a member: a word, proverb, oral history, observation, document, heritage site or natural heritage record
Demonstrator	The working version of this platform that Kakebe has already built
Hypercare	Intensive support in the first period after launch
Idempotent upload	An upload that can be retried safely: the server stores it only once
Leb Lango	The Lango language
Offline first	Designed to work without an internet connection and to catch up when one is available
Outbox	The queue on the phone where contributions wait until they can be sent
PWA (progressive web app)	A website that can be installed on a phone like an app
Push notification	A message that appears on a phone even when the app is closed
Staging	A private copy of the site used for testing and review before changes go public
Tracking code	A short code that lets a contributor follow a submission's status
Won Nyaci	The title of the paramount traditional leader of the Lango

Kakebe

Let's build it in Lango, for Lango.

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A youth-led tech hub building innovators across Northern Uganda.